

2024

Test Rapport GO-e

Meten en analyseren van het slim laden proces van elektrische trucks en DC laadstations



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Dit project is uitgevoerd met Topsector Energie Subsidie van het Ministerie van Economische Zaken en Klimaat, uitgevoerd door Rijksdienst voor Ondernemend Nederland. De specifieke subsidie voor dit project betreft MOOI-subsidie ronde 2020.

1. Introductie

In het volgende document wordt ingegaan op de werkzaamheden die het Elaad Testlab binnen het werkpakket 1b.4 van GO-e heeft uitgevoerd; het onderzoeken van het niveau van ‘smart charging readiness’ van heavy duty laadinfra. Het onderzoek is uitgevoerd door trucks en laadstations te testen op correcte omgang met OCPP Smartcharge profielen en hun invloed op power quality op verschillende vermogens. Het goed functioneren van smart charging en een goede power quality is van belang om de smart charge dienst ontworpen in werkpakket 1b.3 succesvol uit te voeren. In dit project ligt de focus op DC laden met de CCS2 stekker, omdat heavy duty vervoer veelal wordt opgeladen door middel van DC snellaadstations. De testen zijn uitgevoerd in het ElaadNL Testlab. Tijdens de projectduur is ElaadNL verhuisd naar een grotere locatie op industriepark Kleefse Waard te Arnhem. Op deze locatie is geïnvesteerd in een hoog vermogen testsysteem waarmee DC laders en voertuigen getest kunnen worden. Dit systeem is deels gefinancierd met het hardware budget van GO-e.

In de volgende hoofdstukken zal eerst worden ingegaan op de DC laadstation testen. Eerst wordt de testopstelling toegelicht en daarop volgen het test protocol en de resultaten van de smart charging en power quality testen. Vervolgens worden de testen met trucks toegelicht en wordt afgesloten met een conclusie met aanbevelingen.

2. Testen van heavy duty laadinfrastructuur

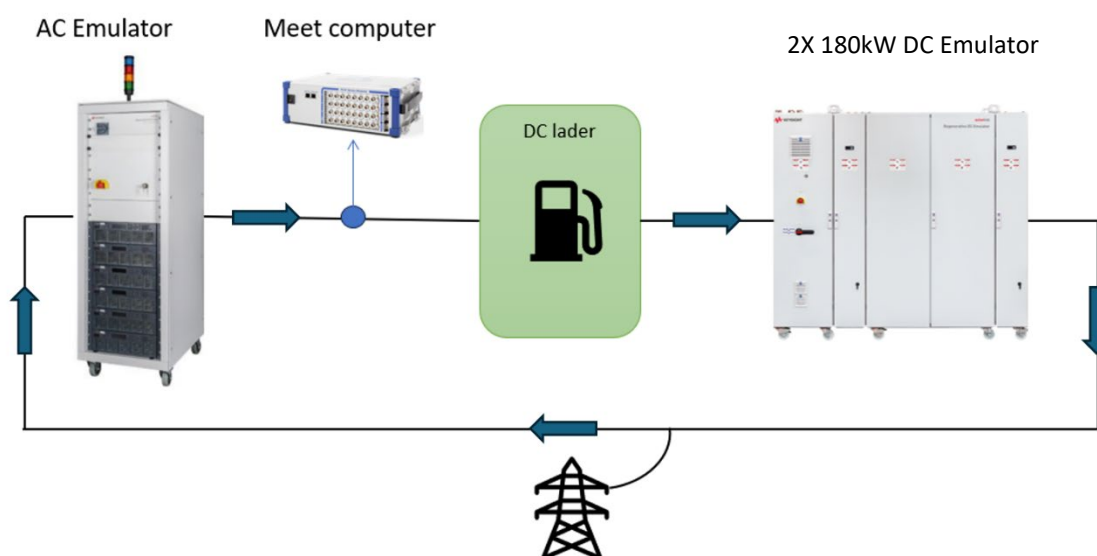
Binnen het GO-e werkpakket 1b.4 “*Validatie van smart charge dienst in Living Lab*” heeft het ElaadNL Testlab het niveau van ‘smart charging readiness’ van heavy duty laadinfrastructuur onderzocht. Het gaat hierbij om laadstations welke op hoog vermogen gelijkstroom kunnen leveren rechtstreeks aan de batterijen van de elektrische voertuigen. Dergelijke laadstations worden ook wel DC laadstations genoemd. DC staat voor Direct Current, ofwel gelijkstroom. ElaadNL heeft 5 van deze DC laadstations getest waarvan de laadsnelheden varieerden tussen de 40 en 360 kW. Het gemiddelde vermogen van de DC laadstations was 188 kW.

De testen bestonden uit smart charging scenario testen, waarin verschillende toepassingen van smart charging zijn getest op de DC laders. Tijdens deze testen werd ook de Power Quality gemeten. In de volgende deelhoofdstukken wordt de testopstelling, het testplan en de testresultaten van de smart charging en power quality testen toegelicht.

Test opstelling

Voor de smart charge- en power quality-testen is een testopstelling van de leverancier Keysight aangeschaft, geïnstalleerd en gebruikt. De testopstelling is weergegeven in figuur 1. De laders worden niet rechtstreeks gevoed vanuit het elektriciteitsnet, maar met een zogenoemde AC-emulator van Keysight. AC staat voor Alternating Current, ofwel wisselstroom. Dit zorgt ervoor dat de lader gevoed wordt met een schoon net, waardoor duidelijk wordt dat eventuele gemeten verstoringen te herleiden zijn tot het laadstation en niet tot een ander apparaat binnen hetzelfde net. De verstoringen worden met een meetcomputer aan de net (AC)-zijde van de lader gemeten.

Vervolgens moet het DC vermogen van de lader ergens heen. Een mogelijkheid is om een voertuig te laden met dit vermogen, maar dit heeft enkele grote nadelen. Namelijk dat het voertuig snel vol is omdat er met een hoog vermogen geladen wordt, het maximale vermogen van het voertuig ontoereikend kan zijn om de DC lader op zijn maximale vermogen te testen, en dat het voltage van de batterij niet gecontroleerd gevarieerd kan worden. Dit resulteert erin dat de testopstelling niet elke keer hetzelfde is en de DC lader niet goed op zijn gedrag op verschillende vermogens en voltages kan worden getest. Dit is niet bevorderlijk voor het onderzoek. Daarom is er getest met een DC Emulator, hiervoor zijn twee Keysight SL1830A Regenerative DC Emulator van elk 180 kW. Dit apparaat zet de DC-stroom die uit de lader komt om in AC-stroom en zet dit terug op het elektriciteitsnet. Hierdoor kan er op hoog vermogen getest worden zonder dat er veel energie verbruikt wordt, omdat de energie terug wordt gevoed op het net. Eén van de twee DC Emulatoren is enkel gebruikt voor het Go-E project, een extra DC emulator was nodig voor dit project om de smart charge readiness van Heavy Duty laadinfrastructuur te testen. Zonder deze extra DC Emulator kon er niet op de beoogde vermogen van 360 kW getest worden.



Figuur 1: Keysight Testopstelling smart charge en power quality testen.

Smart charging readiness testen

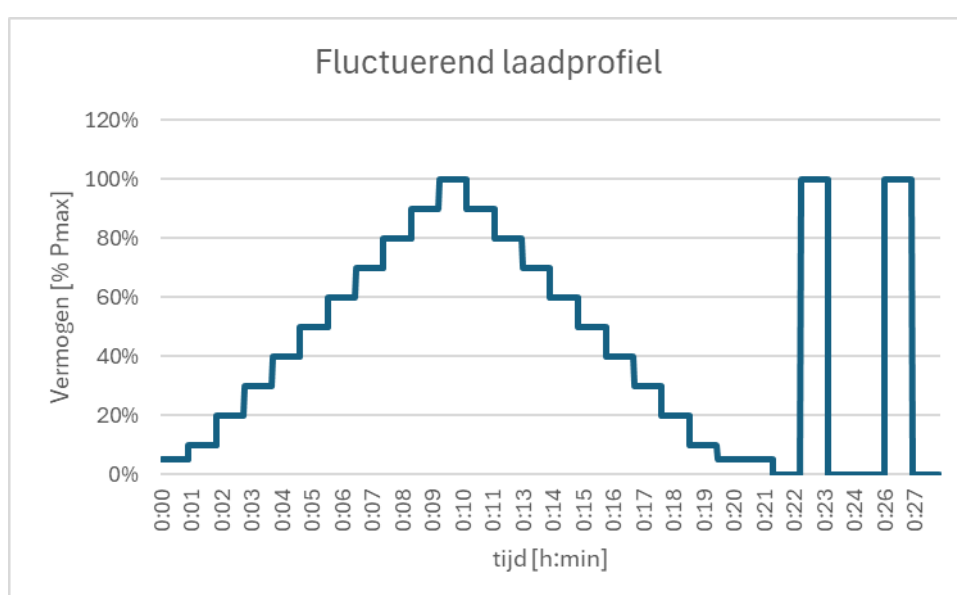
In de volgende secties worden het Smart charging readiness test protocol en de resultaten van deze testen toegelicht.

Test Protocol

Om het slimme laadgedrag van laadstations te onderzoeken is de omgang van de laders met verschillende smart charging scenario's getest. Deze scenario's zijn gemaakt op basis van verschillende toepassingsmanieren van smart charging. Hiervan zijn zogenaamde laadprofielen gemaakt en deze profielen zijn via het door ElaadNL gebruikte backoffice systeem naar de laadstations gestuurd. Tijdens deze testen is er gecontroleerd of de lader daadwerkelijk ging laden met de gevraagde laadsnelheden.

Fluctuerend laadprofiel

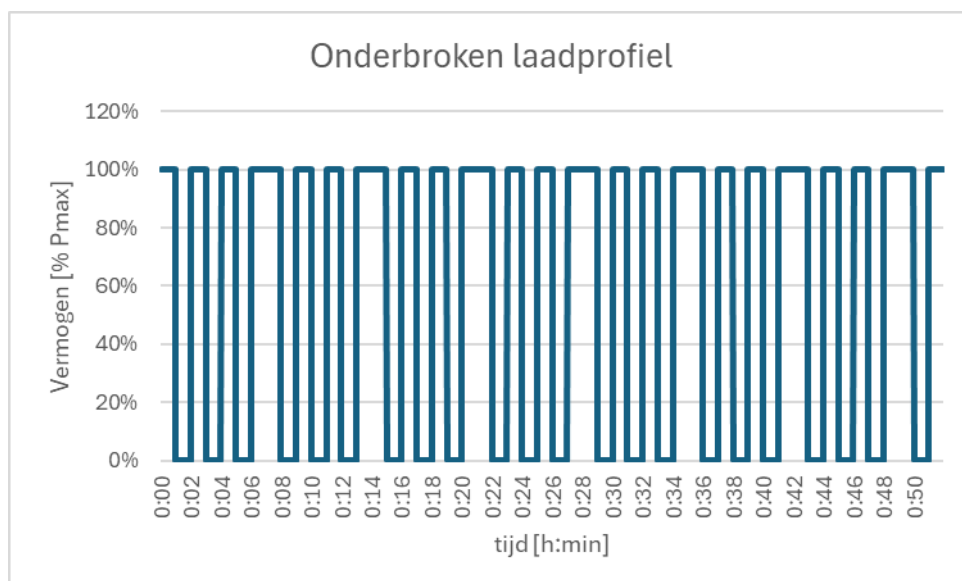
Bij dit laadprofiel wordt het laadgedrag van de lader op verschillende vermogens getest. Van 5% van het maximale vermogen van de lader tot 100% in stapjes van 10%. Als de 100% is bereikt, wordt de lader weer in stapjes van 10% terug geregeld naar 0%. Daarna wordt nog twee keer 100% gevraagd. Dit wordt gedaan om te onderzoeken wat de lader doet als 0% wordt gevraagd en om wat de lader doet wanneer er tussen 0% en 100% gewisseld wordt. Figuur 2 toont het laadprofiel. Tijdens de testen is er gekeken naar de conformiteit tussen het gevraagde en het daadwerkelijke laadvermogen en de reactietijd tussen het verzenden en het aanpassen van het laadvermogen. Tijdens deze test wordt ook de power quality gemeten op de verschillende vermogens. Dit wordt verder toegelicht in hoofdstuk 3.



Figuur 2: Fluctuerend laadprofiel.

Onderbroken laadprofiel

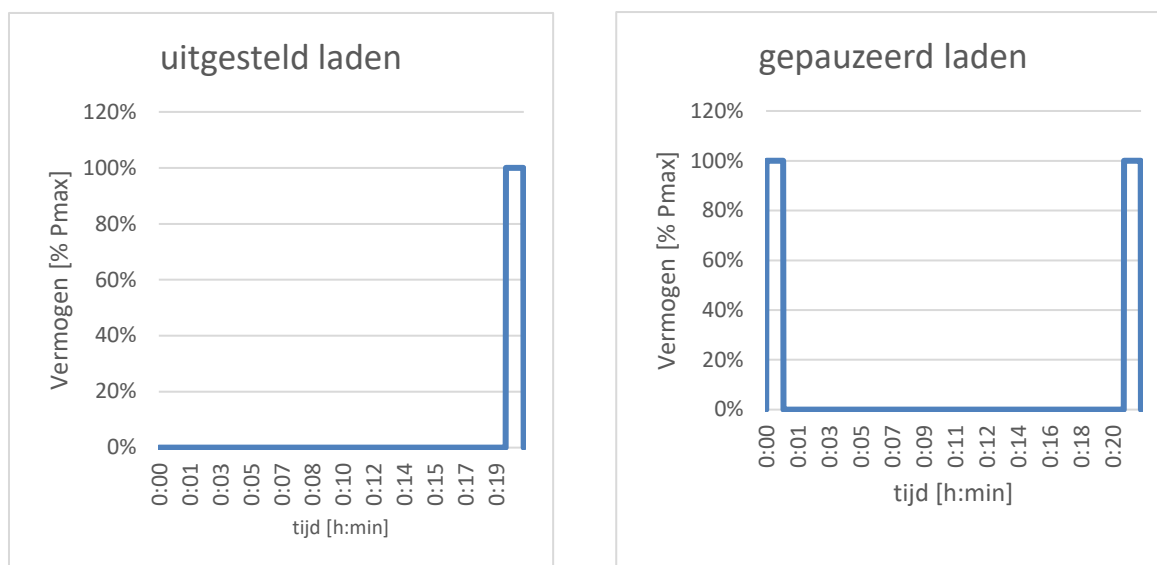
Bij dit laadprofiel is getest of de laadsessie meerder malen gepauzeerd kan worden. Het is van belang dat dit werkt omdat het een andere manier is om met beperkt vermogen meerdere voertuigen te laden. Het laden wordt dan verdeeld over tijdsloten. Voordeel van deze werkwijze ten opzichte van het verlagen van het laadvermogen is dat het vaak energie efficiënter is om korter op een hoger vermogen te laden dan langer op een laag vermogen. Het profiel dat getest, kan teruggevonden worden in Figuur 3.



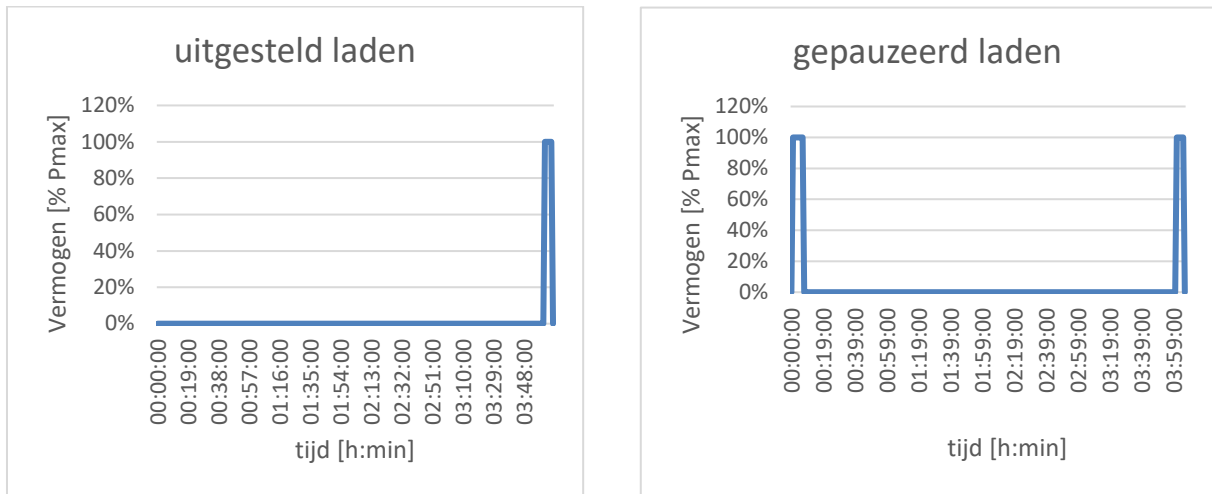
Figuur 3: Onderbroken laadprofiel

Uitgesteld & gepauzeerd laden

Bij het uitgestelde laden is er getest of er voor een bepaalde tijd een laadsessie uitgesteld en gepauzeerd kan worden. Dit wordt getest voor een onderbreking van 20 minuten en van 4 uur. Het verschil tussen uitgesteld laden en gepauzeerd laden is dat er bij uitgesteld laden gewacht wordt met het laden van een voertuig en bij gepauzeerd wordt het laden onderbroken voor een bepaalde tijd. De 20 minuten test heeft betrekking op het onderbroken laadprofiel; vaak zijn de pauzeperiodes in deze rond de 15 minuten. De 4 uur test heeft betrekking op het uitstellen/pauzeren van de laadsessie over piekmomenten in verbruik heen, zoals de avondpiek in woonwijken. De profielen kunnen hieronder gevonden worden.



Figuur 4: 20 minuten uitgesteld en gepauzeerd laden



Figuur 5: 4 uur uitgesteld en gepauzeerd laden

Resultaten

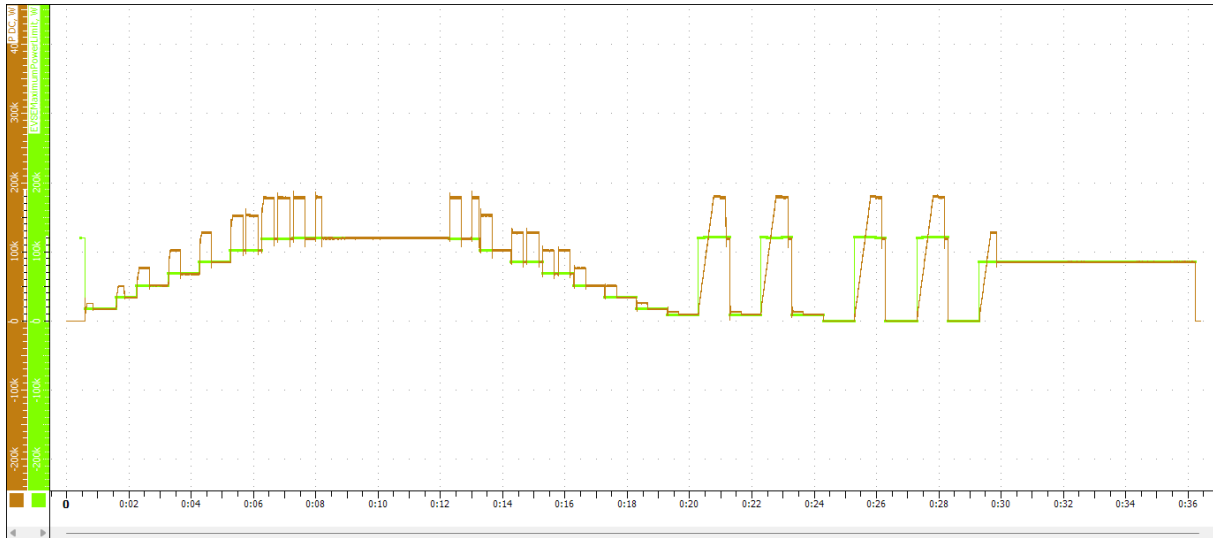
De hierboven genoemde testen werden uitgevoerd op vijf DC-laders. De resultaten van dit onderzoek zijn te vinden in tabel 1. In dit onderzoek zullen alleen geaggregeerde en geanonimiseerde resultaten worden genoemd. Dit omdat het Elaad Testlab contracten heeft met de fabrikanten van de laders waarin is vastgelegd dat er geen resultaten gedeeld worden met derden die naar hun product herleid kunnen worden.

Tabel 1: Geaggregeerde resultaten van het onderzoek naar het slimme laadgedrag van DC-laders.

Profiel	goed	goed met opmerkingen	niet goed	niet ondersteund
Fluctuerend laadprofiel	2	1	1	1
Onderbroken laadprofiel	0	0	0	5
Uitgesteld en gepauzeerd laden	0	0	0	5

Fluctuerend laadprofiel

De resultaten van de testen over het fluctuerend laadprofiel waren als volgt. Twee laadstations presteerden goed en voldeden aan dit profiel, dat wil zeggen dat het vermogen van het laadstation op de gevraagde waarden bleef en snel genoeg reageerde. Eén laadstation functioneerde over het algemeen goed, maar overschreed lichtjes het beoogde vermogen. Een ander laadstation volgde weliswaar een laadprofiel, maar ging aanzienlijk boven het aangegeven laadvermogen. Dit resultaat wordt weergegeven in 6, waarin de groene lijn het gevraagde vermogen is en de bruine het daadwerkelijk geleverde vermogen weergeeft. Het overschrijden van dit vermogen kan problemen bij een laadplein veroorzaken, zoals het overschrijden van de maximale stroomwaarden. Dit kan weer leiden tot het afschakelen van het hele laadplein doordat de hoofdzekering afschakelt. Tot slot bleek dat één lader de smart charging module niet had geïmplementeerd in hun OCPP-implementatie, hierdoor was het helemaal niet mogelijk om smart charging toe te passen.



figuur 6: Overschrijding van het gevraagde vermogen.

Onderbroken laadprofiel & uitgesteld en gepauzeerd laden

Onderbroken, uitgesteld en gepauzeerd laden tijdens DC laadsessies wordt alleen ondersteund door laadstations die communicatie via het ISO 15118-20 protocol ondersteunen. De heavy duty DC laadstations die tijdens de projectduur in het Elaad Testlab aanwezig waren ondersteunden alleen DIN SPEC 70121 en ISO 15118-2, zij ondersteunen dus niet onderbroken, uitgesteld en gepauzeerd laden. Het ontbreken van laadstations die ISO 15118-20 spreken komt omdat deze standaard pas eind 2023 is uitgekomen. Hierdoor is er nog maar een handje vol DC laadstations die dit wel kunnen en deze hebben hun weg naar het Elaad Testlab nog niet gevonden.

3. Power Quality

Power quality (PQ) is een verzamelterm voor verschillende verstoringen in het elektriciteitsnet. Deze verstoringen ontstaan bijvoorbeeld door het omvormen van energie van AC naar DC. Bij DC laden is het laadstation verantwoordelijk voor de PQ van de laadsessie, omdat de omvorming van AC naar DC gebeurt in het laadstation. Een DC laadstation kan een slechte PQ veroorzaken, maar een laadstation kan ook last hebben van een slechte PQ. Het is daarom van belang dat de laadstations een beperkte invloed op de PQ hebben.

Test protocol

Tijdens het testen van het fluctuerende laadprofiel wordt ook de power quality van de lader gemeten. Deze meting gebeurt aan de Keysight AC emulator (netzijde) van de lader, zoals afgebeeld in figuur 1. Voor dit onderzoek hebben we vooral gekeken naar de power factor (PF) die de laadpaal heeft tijdens het laden. De PF geeft aan hoeveel van de getransporteerde stroom wordt omgezet in bruikbare energie, in dit geval voor het laden van een voertuig. Bij een PF van 1 wordt alle energie omgezet in bruikbare energie. Bij een PF van 0 wordt de stroom niet omgezet in bruikbare energie, echter moet deze energie alsnog door de netbeheerder vervoerd worden. Volgens de huidige

netcode is een PF lager dan 0,85 niet toegestaan. In dit onderzoek kijken we naar de PF over de hele range van de laadpaal met het profiel beschreven in figuur 2.

Naast het onderzoek over de PF is er een afstudeeropdracht uitgevoerd door een masterstudent over de (supra-)harmonische verstoringen van de laders. Dit onderzoek kan gevonden worden in appendix A.

Geaggregeerde resultaten

De resultaten van de gemeten PF kan gevonden worden in tabel 2. Alleen op de laadstations die smart charging ondersteunen is de PF getest. Zichtbaar is dat de 4 laders op vol vermogen een goede PF hebben, dat wil zeggen boven de 0.85. Echter, wanneer het laadvermogen naar het minimale vermogen van het laadstation gaat voldoen 3 van de 4 laadstations niet meer aan de standaarden. Gemiddeld hebben de laadstations dan een PF van 0.37 met een gemiddeld actief vermogen van 2.1 kW en schijnbaar vermogen van 8.2 kW.

Tabel 2: Resultaten PF van DC Laders

	Goed	Niet goed
PF op maximale vermogen van de lader	4	0
PF op het minimale vermogen van de lader	1	3

4. Testen van Truck

Test protocol

Tot op heden zijn er in het Elaad Testlab twee heavy duty trucks van 1 fabrikant getest. Deze trucks zijn getest volgens het vehicle test protocol. Dit protocol is te vinden in appendix B. Tijdens deze testen worden dezelfde slimme laadprofielen gebruikt die gebruikt zijn in hoofdstuk 1.

Daarnaast is de interoperabiliteit getest door de smart charging mogelijkheden van de trucks te testen in combinatie met de laadstations die voor dit onderzoek zijn gebruikt.

In het projectplan werd tevens het volgende voorgesteld: *Om slim laden goed uit te kunnen voeren is het van belang dat van elk voertuig bekend is wat zijn laad curve is; door een volledige laadcyclis te doorlopen en de energieopname te meten kan worden bepaald bij welke State of Charge een voertuig met welke laadsnelheid kan laden. Voor een goed resultaat dient dit voor een aantal maximum laadsnelheden gemeten te worden, aangezien de snelheid van laden ook weer invloed heeft op de batterijtemperatuur, en daarmee de laadcurve.* Deze testen waren voor Elaad niet uitvoerbaar omdat voor het testen van een laadcurve het voertuig (bijna) leeg moet zijn en na de test het voertuig vol is. ElaadNL heeft de mogelijkheden niet om het voertuig te ontladen of leeg te rijden. Dergelijke testen zijn beter uitvoerbaar in de praktijk, waar het voertuig na laden en registreren van de laadcurve ook weer nuttig kan worden ingezet.

Resultaten

Omdat er tot op heden slechts 2 trucks getest zijn bij het Elaad Testlab, is het niet mogelijk om geaggregeerde resultaten te delen. Vanwege het lage aantal, en aangezien het Elaad Testlab een open lab is met veel bezoekers, is niet te garanderen dat de resultaten niet herleidbaar zijn naar de fabrikant en zijn individuele producten.

Wel zijn de resultaten van deze testen gedeeld met de fabrikant van het voertuig. Zij gaan met de bevindingen en aanbevelingen aan de slag om het product te verbeteren.

5. Conclusie

Uit de uitgevoerde testen zijn enkele belangrijke bevindingen naar voren gekomen met betrekking tot de prestaties van de laadstations. Van de vijf geteste laadstations ondersteunde één het Smart Charge-gedeelte van het Open Charge Point Protocol (OCPP) niet. Twee laadstations waren in staat om op een correcte manier te smart chargen, terwijl één laadstation wel kon smart chargen maar dit deed met een iets te hoge snelheid. Daarnaast laadde een ander laadstation met een veel te hoge snelheid ondanks dat het smart charging ondersteunde. Op vol vermogen bleek de Power Factor (PF) voor elk laadstation goed te zijn, maar op minimaal vermogen was de PF slechts bij één van de vier laadstations goed.

Deze bevindingen leiden tot de volgende adviezen. Aan fabrikanten adviseren we om ervoor te zorgen dat het Smart Charging-gedeelte van het OCPP protocol volledig geïmplementeerd is in hun laadstations. Tevens raden we aan om hun voertuigen of laadstations bij het Elaad Testlab te laten testen om een juiste implementatie van smart charging en een goede power quality te garanderen.

Voor laadpaalexploitanten en -eigenaren is het van belang om vóór aanschaf van een laadstation of voertuig te vragen of het product getest is bij het Elaad Testlab en wat de bevindingen van deze testen waren. Fabrikanten zijn vrij om het door het Elaad Testlab opgestelde rapport met hun klanten te delen. Het Elaad Testlab mag dit juridisch niet zelf doen. Het Elaad Testlab is voornemens om alle aangeboden trucks en DC laadstations te blijven testen op smart charging, interoperabiliteit en power quality. Daarnaast voert het Elaad Testlab cyber security scans op de laadstations en voertuigen uit. Deze vallen buiten de scope van het GO-e project, maar worden nog wel uitgevoerd op de geteste laadstations.

Deze punten en adviezen onderstrepen het belang van zorgvuldige selectie en testen van laadstations om een optimale werking te waarborgen. Het implementeren van de aanbevelingen draagt bij aan een betrouwbare en goed met het elektriciteitsnet geïntegreerde laadinfrastructuur, wat essentieel is voor de verdere ontwikkeling van elektrisch vervoer.

Analysis of Emitted Electromagnetic Interference of DC Electric Vehicle Supply Equipment

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Abstract—High-power electric vehicle chargers are being installed on a large scale. These devices use converters that convert AC power to DC power. They generate distortion in the supraharmmonic spectrum which can cause varying issues. This research measured and characterized the emission of three high-power electric vehicle chargers under different operating conditions using various analysis methods. The chargers are also tested against proposed emission limits. The results varied significantly between EVSEs. Results show a distortion's amplitude and frequency can change over time. These findings can be used when developing feasible standards for the regulatory gap.

Index Terms—EMI, power quality, supraharmmonic distortion, supraharmmonic current, DC EVSE, operating conditions, time-frequency analysis

I. INTRODUCTION

Transitioning from fossil fuels to renewable energy sources poses many challenges. One of these challenges is the electrification of the transport sector, which is achieved by replacing internal combustion engine vehicles with electric vehicles (EVs). A common reason for consumers against purchasing EVs is the limited range. This issue can be solved by installing DC electric vehicle supply equipment (EVSE) along highways. These devices can reach higher charging speeds than conventional chargers, allowing drivers to increase their range significantly in a 15-minute break. The difference between conventional and fast charging is that conventional charging uses the onboard charger (OBC) in the EV to convert the AC grid power to DC, while fast charging uses a DC EVSE to perform the conversion. The OBC is located in the car, which limits its weight and size. DC EVSEs do not have this restriction, allowing them to charge at over 300 kW. A fast-charging plaza with tens of these devices can easily draw several megawatts at peak consumption. This threatens the electricity grid's power quality as converters are notorious for generating distortion in the supraharmmonic (SH) range [1].

Power quality rules regulate electromagnetic interference (EMI) below 2 kHz, while electromagnetic compatibility (EMC) regulations apply to frequencies above 150 kHz. This leaves a regulatory gap between 2 kHz and 150 kHz, known as the SH frequency range [2]. Although the name "supraharmmonic" suggests it concerns distortion at integer multiples of the grid

frequency, the SH frequency range is a continuous spectrum. Though the spectrum is not regulated yet, supraharmmonic distortion (SHD) can cause problems such as unwanted tripping of earth leakage current breakers, failure of measurement instruments, audible noise, and undesired switching of lighting [3] [4]. Because of these effects, proper standards must be devised to limit SHD.

Currently, EMC compliance testing is often conducted using an EMI receiver. This device is connected to the device under test (DUT) and measures its distortion. The distortion is analyzed in the frequency domain by applying an analog bandpass filter and measuring the total emission in the band. The size of these bands is called the resolution bandwidth (RBW). The RBW changes depending on the frequency. For example, it could be 200 Hz below 9 kHz and 5 MHz between 30 and 300 MHz. The EMI receiver sweeps through the spectrum and measures one band at a time. A few quantities are determined during a measurement, such as peak and average distortion. Measuring all bands creates a graph of these quantities over frequency. These graphs are compared to limits to determine if the DUT is compliant. Each band is measured consecutively, so one sweep through the spectrum is time-consuming. Since the DUT may have hundreds of different settings and operating conditions that all have to be tested, the compliance test can take a long time. The specific procedure and all variables such as the RBW and the limits of the EMI are defined in standards like the IEC61000-4-7[5]. These standardizations allow for a fair comparison between DUTs, but that does not imply that the test environments described in the standards represent realistic scenarios. The standards for the SH range must be designed to provide fair and repeatable results. This can be achieved by analyzing the behavior of SHD to determine which behaviors the tests need to account for. The limits must be set at an achievable level for manufacturers, but also such that few EMC issues arise.

In [6], current emission limits, also known as the "Slangen limits", are proposed for the SH range on an installation level. This means they apply to the total emission of, for example, a charging plaza. The limits are derived from research on realistic grid impedances and the IEC 61000-2-2, which states voltage

compatibility limits. The limits can vary based on the grid impedance at the connection point of the installation. To not exceed the installation limit when connecting multiple devices, individual devices are recommended to remain below 25% of the installation limit.

Several ways to analyze SHD exist, such as supraharmonic current (SHC), total supraharmonic current (TSHC), supraharmonic voltage (SHV), and total supraharmonic voltage (TSHV). In [7], the SHCs of an EVSE for several operating conditions are measured. They found a strong correlation between emitted SHC and the DC output voltage of the EVSE. The output power did not affect the emitted SHC.

In [8], the SHD of various OBCs in frequency and time domain are characterized. They analyzed the voltage distortion from 9 kHz to 500 kHz. They found the amplitude of the distortion was time-dependent and repeated itself over time.

This research aims to provide insight into SHD's characteristics at the grid side of individual high-power DC EVSEs. This knowledge can contribute to the division of standards for the SH frequency range. Variations in time, operating condition, and resolution bandwidth, i.e. bin size are analyzed. Variations in operating conditions are also compared between multiple EVSEs. The EVSEs are tested against proposed limits to assess if they are realistic for high-power EVSEs. While not the focus, interesting phenomena outside of the SH range will also be mentioned.

In Section II, relevant concepts for supraharmonic analysis are explained. It also describes the conducted measurements and their motivation. Section III provides an overview of the measurement setup and lists the DUTs specifications. Section IV shows the collected spectra, spectrograms, and distortion per operating condition graphs. The graphs and their implications are described, as well. In Section V, conclusions from the measurement data are stated. Section VI lists the author's recommendations for future work.

II. ANALYSIS

This section lists and explains what measurements are conducted and why certain choices are made. It also expands on SHC, TSHC, and the influence of grid impedance on current distortion.

A. SHC and TSHC

The SHC is analogous to the RBW of EMI receivers. Because of the analogy, some conclusions regarding the SHC can also be applied to the RBW. The SHC is described in Equation 1. It is the root sum square of all frequency components in the bandwidth between f_{min} and f_{max} . Choosing these values close to each other creates a bin with relatively low noise. However, if a bin is too small, a distortion with time-varying frequency could sometimes end up outside the bin. This leads to inconsistent

results. Increasing the bin size increases the noise contribution, but decreases the likelihood of a distortion peak leaving the band. Another downside of a larger bin is that it increases the probability of multiple distortion peaks being in the bin. This is undesirable for the analysis of individual distortion peaks.

$$I_{SH} = \sqrt{\sum_{f=f_{min}}^{f_{max}} I_f^2} \quad (1)$$

Calculating the SHC with $f_{min} = 2$ kHz and $f_{max} = 150$ kHz results in the total supraharmonic current (TSHC). This quantity shows the total current in the SH spectrum. All frequencies contribute equally to the final value. Higher frequencies often have lower emission limits in standardization such as, for example, the Slangen limits. The reason is that higher frequencies tend to have more impact at equal magnitude than lower frequencies. Hence, the TSHC poorly summarizes an EVSE's impact because higher frequency distortions are under-represented.

B. Grid impedance

Grid impedance plays a vital role when measuring distortion. This is illustrated using Figure 1. It shows a simplified model of a 1-phase grid with an appliance modeled as a current distortion source and an impedance. The measured current distortion, i.e. the high-frequency current through Z_{grid} , decreases if the grid impedance increases. However, this does not mean a higher grid impedance is better, because it causes an increase in voltage distortion.

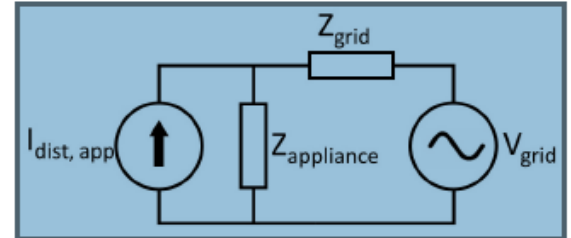


Fig. 1: Simplified model of a 1-phase grid with an appliance connected.

Because the grid impedance affects the measured current, comparing absolute values of SHC is only possible if the same grid impedance is used for each measurement. Usually, this is achieved by using a line impedance stabilization network (LISN). This device measures the current indirectly by measuring voltage over a predefined impedance. However, a LISN uses a lowpass filter to separate the distortion from the power-carrying frequency. Hence, a LISN works best at high-frequency distortion (150+ kHz). A different option is to measure current directly using current transducers. These devices work better at lower frequencies, compared to the LISN. Hence, they are more suitable for this research.

C. Frequency spectra

Frequency spectra of all EVSEs are used to compare the distortion characteristics of the EVSEs. The spectra are calculated by applying a 200 ms rectangular window to the center of a 500 ms measurement, after which the Fourier transform is calculated. The rectangular window does not attenuate the signal, but it does not prevent spectral leakage from occurring either. A 200 ms window results in a frequency resolution of 5 Hz after the Fourier transform is calculated, assuming a 1 MHz sampling frequency. The grid frequency must be divisible by the frequency resolution because distortion is expected at multiples of the grid frequency. A random frequency resolution would split these distortions over two frequency components, attenuating the magnitudes. The spectra's characteristics are anticipated to vary between EVSEs because they have different manufacturers that likely used different converters and EMI filters. The spectra are compared to the Slangen limits. A lower grid impedance allows higher limits, but no relation between the limit and the grid impedance is provided. Hence, 25% of the worst-case scenario of the installation limit is used as the individual device limit.

D. SHC characteristics at various operating conditions

The SHCs of 3 EVSEs are calculated for various DC currents and voltages to compare to the results found in [7]. The results are expected to be different since the EVSEs have different manufacturers and thus different converters and EMI filters. The measurements are performed as follows. Based on the specifications of the EVSE, a table with possible operating conditions is generated. Each operating condition is a unique combination of DC voltage and current. An example of the operating condition table is shown in Table I. Note that not all combinations of voltage and current are possible due to the EVSE's power limit.

TABLE I: Example of each operating condition's power for a 250 A, 1000 V, 180 kW EVSE. Conditions in red exceed this EVSE's power limit and can not be created.

DC Voltage (V)	DC Power (kW)		DC Current (A)			
	250	300	0	50	...	200 250
...	...	...	0	12.5	...	50 62.5
700	0	0	0	15	...	60 75
750	...	...	...	...	...	...
800	0	0	0	35	...	140 175
850	0	0	0	37.5	...	150 187.5
900	0	0	0	40	...	160 200
950	0	0	0	42.5	...	170 212.5
1000	0	0	0	45	...	180 225
	0	0	0	47.5	...	190 237.5
	0	0	0	50	...	200 250

A 500 ms measurement is performed at each operating condition. The Fourier transform is calculated in the same way as stated above. Then, a bin around a dominant frequency component is selected to calculate the SHC. Since distortions could vary in frequency over time, the bin size is selected manually to ensure the peak remains in the bin. The bin sizes

for EVSE 2 and 3 are determined recursively. The starting size depends on the presence of nearby distortions. For example, if the frequency to be analyzed is at 3 kHz, and distortion occurs at 2.95, 3, and 3.05 kHz, the starting bin size will be 50 Hz. If there is more space, the starting bin size is 200 Hz. The size is increased if the data shows that all measurements after a certain time show a significantly lower SHC, which is likely caused by the distortion leaving the bin. A sufficient bin size is reached if the significant drop in SHC is no longer present. If the significant drop does not vary over time when increasing the bin size, it is likely due to the charger's behavior, and a coincidence that the last measurements have significantly lower SHC. For example, an EVSE has 4 operating conditions that are measured consecutively. If the SHCs for the operating conditions are 35, 40, 5, and 3 for a 200 Hz bin, and 36, 41, 31, and 33 for a 500 Hz bin, the distortion probably left the 200 Hz bin between the second and third measurements.

E. Spectrogram

A 1-hour test where EVSE 1 supplies 650 V and 150 A is conducted to produce a spectrogram to examine slow spectral time variation. These variations can lead to time-dependent compatibility results. For example, the EVSE may pass the test in the first 5 minutes, but exceed the limits afterward. The test starts in the morning when the EVSE has not heated up from other tests. This is done so that potential temperature-related variation can be observed. During the test, a 500 ms measurement is performed every 20 seconds, resulting in 180 measurements. The spectra of all measurements are calculated identically to the way stated above. The spectrogram is produced by plotting all spectra in a 3D graph in chronological order.

III. MEASUREMENT SETUP

This section describes the measurement setup, what equipment is used, and the specifications of the DUTs.

The measurement setup is illustrated in Figure 2. A script controls the charging discovery system (CDS) and the test computer. The CDS emulates a car starting a charging session and requests the EVSE for a certain voltage and current. The DC output current of the EVSE is not used to charge a car or battery, since these may contain uncontrollable variables like maximum charging current or the battery's state of charge. Instead, the power is fed to a DC emulator which converts the DC power to 3-phase AC power and feeds it back to the electricity grid. The AC grid emulators create a low-distortion 3-phase grid that powers the EVSE. A set of 5 Danisense DN1000ID current transducers is used to measure the currents in L1, L2, L3, N, and the PE cables. The transduced currents are sampled using a 24-bit 1 MS/s sample rate analog-to-digital converter with built-in shunts, which is part of the test computer. Unfortunately, only a test computer with a 16-bit 1 MS/s sample rate analog-to-digital converter was available for the measurements of EVSE 2. Hence, the noise floor in its spectra may be raised. The

specifications of all EVSEs can be found in Table II. The grid impedance of each phase to neutral at the EVSE's connection point is shown in Figure 17 in Appendix A.

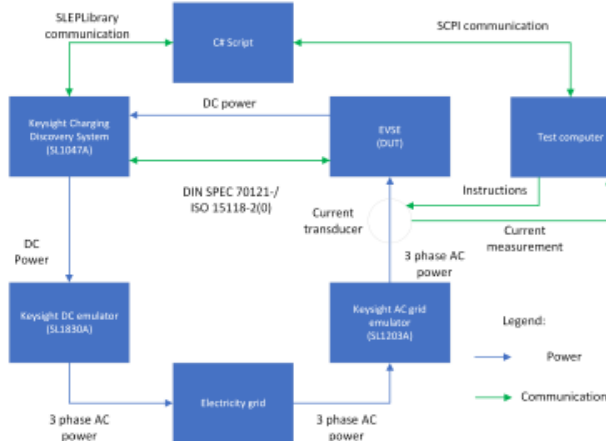


Fig. 2: Schematic overview of the measurement setup.

TABLE II: Specifications of the measured EVSEs.

	Voltage (V)	Max. Current (A)	Max. Power (kW)
EVSE 1	150 - 920	500	350*
EVSE 2	250 - 1000	250	180
EVSE 3	350 - 1000	500	150

*Limited by grid emulators. The practical limit is 330 kW.

IV. RESULTS

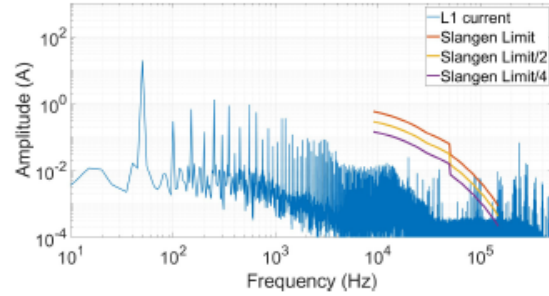
This section presents and discusses the collected measurement data. Each EVSE is discussed in a separate subsection. The results of the EVSEs are compared.

A. EVSE 1

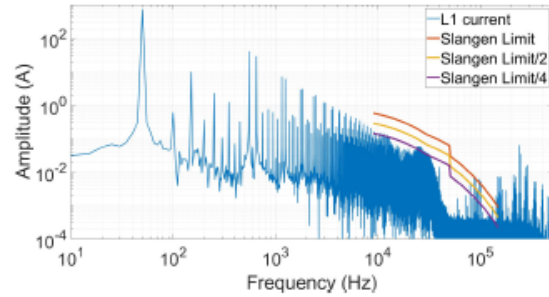
Figure 3a and Figure 3b show EVSE 1's spectra while charging at 650 V, 0 A and 650 V, 500 A, respectively. The 500 A measurement has significantly higher distortion, which is unexpected as it contradicts findings in previous work. The spectrum in Figure 3b between 20 kHz and 40 kHz creates a hump-like shape, while Figure 3a shows little distortion in this range, aside from some narrowband peaks. Both figures show wideband distortion of similar magnitude around 90, 150, 212, 280, 338, 400, and 463 kHz. Figure 3a also shows wideband distortion around 30 kHz. This implies the fundamental distortion is at 30 kHz and only the odd harmonics of this 30 kHz distortion exceed the noise floor, which would result in steps of 60 kHz. The hump-like shape overshadows the 30 kHz distortion in Figure 3b, so it is not visible. Both figures also show a narrowband peak at exactly 240 kHz. Aside from the 150 kHz wideband distortion which exceeds the installation limit, the 0 A spectrum remains below 25% of the Slangen limit. The 500 A spectrum slightly exceeds the 25% limit at 10 kHz

and 20 kHz and, like the 0A spectrum, it exceeds the installation limit at 150 kHz.

Fig. 3: EVSE 1: Frequency spectra.



(a) EVSE 1: Frequency spectrum while supplying 650 V, 0 A.



(b) EVSE 1: Frequency spectrum while supplying 650 V, 500 A.

Figure 4 shows the SHC of EVSE 1 in the 2.95 kHz bin with a bin size of 50 Hz. Constant power curves are included in the figure. The SHC increases both with an increase in DC current and DC Voltage. The SHC decreases at all measurement points around 250 kW. The constant power curves above 100 kW align with the SHC plot, which suggests the distortion is constant for constant power above 100 kW.

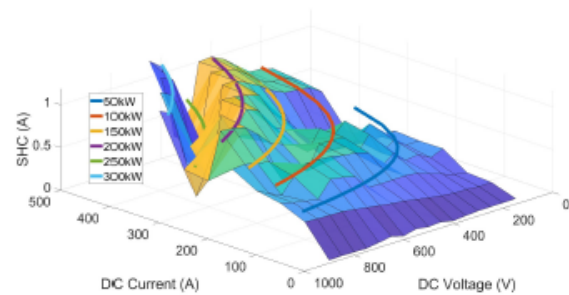


Fig. 4: EVSE 1: SHC in the 2.95 kHz bin and constant power lines.

Figure 5 shows Figure 4's datapoints sorted by power. Three trend lines are identified in red. The top two merge at 140

kW, and the bottom line merges with the others at 290 kW. Aside from the decrease in distortion between 250 and 300 kW, the SHC increases with power. The lines suggest the device's distortion fluctuates between three modes and is not completely random. The data does not show a clear relation between power and SHC in this frequency bin.

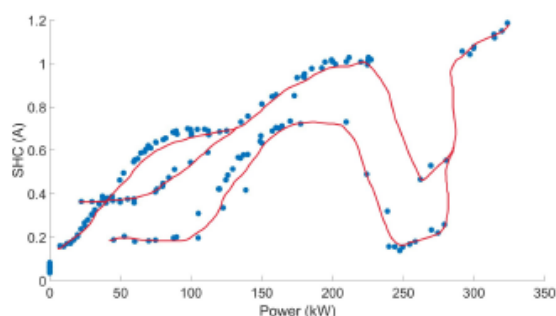


Fig. 5: EVSE 1: SHC in the 2.95 kHz bin against power in blue and trend lines in red.

Figure 6 shows the 24.5 - 26.5 kHz range of the spectrogram which resulted from the 1-hour 650 V, 150 A measurement of EVSE 1. It contains an odd harmonic of the 50 Hz grid frequency every 100 Hz. These are present in the spectrogram between 0 and 48 kHz with decreasing amplitude over frequency. Figure 6 also contains a "traveling" distortion that decreases 2 kHz in frequency during the measurement. The frequency change rate decreases over time. Determining if the amplitude of the traveling distortion is constant is difficult as it crosses the frequency-stable harmonic distortion peaks. The harmonics can cause constructive or destructive interference with the traveling signal, depending on their relative phase.

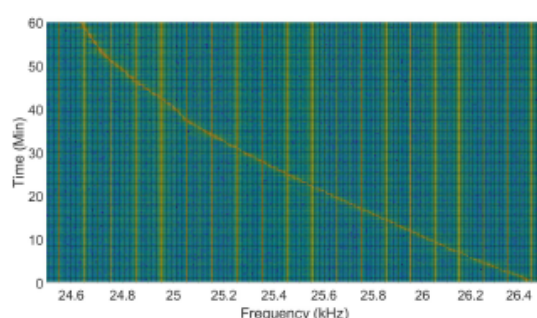


Fig. 6: EVSE 1: Spectrogram from 24.5 to 26.5 kHz while charging at 650 V, 150 A.

Figure 7 shows the 71.0 - 73.0 kHz range. It shows a magnitude-varying distortion peak at exactly 72 kHz. The changes are not repetitive, but rather random.

Figure 8 shows the 87.3 - 89.2 kHz range. It shows a single traveling distortion peak that changes its frequency by 1.7 kHz

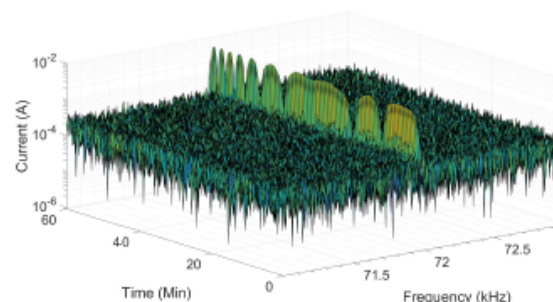


Fig. 7: EVSE 1: Spectrogram from 71.0 to 73.0 kHz while charging at 650V, 150A.

over the 1-hour measurement. The frequency change seems gradual, but a small bump occurs at 40 minutes. This bump is also visible in Figure 6. Although the 88 kHz and 25 kHz traveling peaks have a similar shape, they cannot be some harmonic of the same fundamental. If they were, the frequency change over time of the 88 kHz traveling peak should be 88/25 higher than the frequency change of the 25 kHz peak, which it is not.

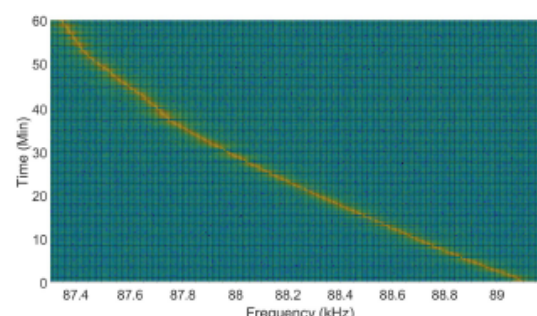


Fig. 8: EVSE 1: Spectrogram from 83.3 to 89.2 kHz while charging at 650 V, 150 A.

Figure 9 shows the 98.8 - 100.8 kHz range. It contains a traveling peak that increases by 1.9 kHz over the 1-hour measurement. Its shape is again similar to the 88 kHz and 25 kHz traveling peak, including the bump at 40 minutes, but its frequency increases, instead of decreases.

Figure 10 shows EVSE 1's SHC over time for various bin sizes of the distortion shown in Figure 8. Selecting the larger binsize suggests the peak's amplitude is stable, which is correct. Selecting a smaller bin size suggests the magnitude is time-dependent, which is incorrect. Notice that when the distortion is inside the small bin, the larger bin remains higher because of the extra noise components. In compliance testing, static bands are measured consecutively. This means that if the frequency sweep speed of the EMI receiver matches the speed at which the distortion changes frequency, a narrowband distortion like in Figure 9 is observed as a 2 kHz distortion band by the

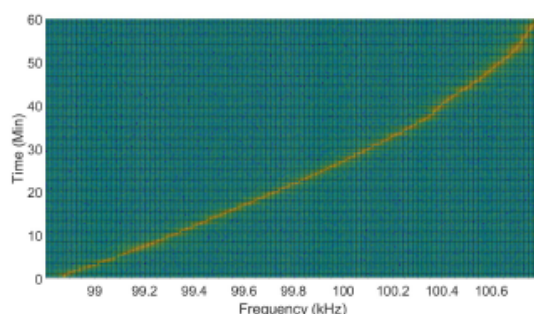


Fig. 9: EVSE 1: Spectrogram from 98.8 to 100.8 kHz while charging at 650V, 150A.

EMI receiver. This is one of many issues that can occur with frequency-varying distortion and consecutive compliance testing. The distortion's frequency can also increase slower than the frequency sweep speed. Or the distortion frequency could decrease. All these scenarios lead to different problems that cause time-dependent compatibility results. Although Figure 10 suggests a sufficiently large RBW solves these problems, this is only true if the band's center frequency is chosen such that all frequencies at which the distortion can occur are inside the band. This is hard to achieve without manually inspecting the data. Hence, it is hard to implement in standardized compliance tests. A solution is to have an overlap between bands. This way, the frequency-varying distortion will always be detected. However, this can still make the distortion appear more wideband than in reality. Besides, introducing overlap increases the compliance test duration, which is undesirable. A different solution is to measure all frequencies simultaneously using, for example, a time-domain EMI receiver. This way, only the observed distortion's frequency can vary, but the amplitude is constant.

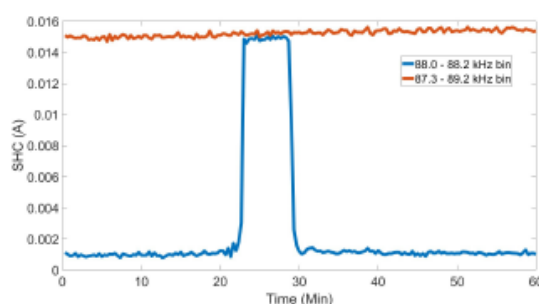


Fig. 10: EVSE 1: L1's SHC over time for varying bin sizes while constantly charging 650 V, 150 A

Figure 11 shows the TSHC of the 1-hour measurement. The TSHC increases by 3% during the test, with the most significant change happening in the first 10 minutes. After 1 hour, the maximum TSHC difference between phases is 1.7%. Although these changes are small, they can make the difference between

passing or failing a compliance test. Hence, more research is required to determine what causes the deviation. Does it only occur if the EVSE is turned on right before the test, or does it also happen if the EVSE has been in standby mode for some time? Do all EVSEs exhibit this behavior? Since the data shows a difference between phases, it is recommended to do compliance testing on all phases.

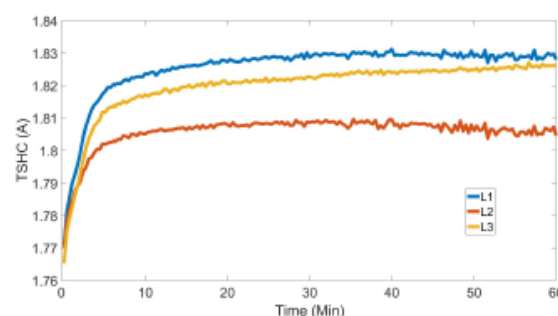


Fig. 11: EVSE 1: TSHC over time.

B. EVSE 2

Figure 12 shows the spectrum of EVSE 2 while charging at 950 V, 150 A. Notice that the noise floor is raised because of the 16-bit analog-to-digital converter. It exceeds the Slangen limits at higher frequencies. The spectrum shows harmonics up to roughly 5 kHz. It also shows narrowband distortion at 22 and 44 kHz. Since the 22 kHz distortion is the highest (aside from harmonic distortion), it is most likely the converter's switching frequency. The 44 kHz distortion is probably the switching frequency's second harmonic. A large distortion is present at 240 kHz. The spectrum also shows wideband distortion at several frequencies starting at 100 kHz. However, they are difficult to analyze because of the raised noise floor. The distortion at 22 kHz, 44 kHz, and 100 kHz exceed the installation level Slangen limit.

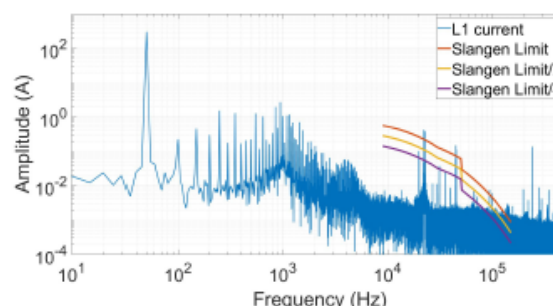


Fig. 12: EVSE 2: Frequency spectrum while charging at 950 V, 150 A.

Figure 13 shows EVSE 2's SHC per operating condition in the 22 kHz bin. The data shows a weak correlation between DC current and SHC.

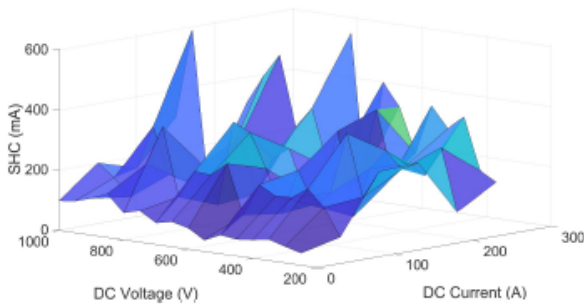


Fig. 13: EVSE 2: SHC in the 22.335 kHz bin.

C. EVSE 3

Figure 14 shows the spectrum of EVSE 3 while charging at 450 V, 0 A. Two distortion shapes are identified. The first is the largest peak at 18 kHz and all its harmonics up to 108 kHz. These distortion ranges have a high middle and decrease with distance from the center of the distortion. The second distortion shape, which has a flat top, is found at 150.3 kHz and every interval of precisely 62.7 kHz afterward. It is possible that this shape would be at 87.6 kHz, but it overlaps with the 5th harmonic of the switching frequency at 90 kHz, so this is unclear. At $87.6 - 62.7 = 24.9$ kHz, another peak is present. This is likely a coincidence. The difference of all peaks is 62.7 kHz, so the fundamental must be either 62.7 or 31.35 kHz. No distortion shows at 62.7 kHz, but there is a narrow peak at 31.35 kHz. However, this peak does not have the same shape as the wideband distortion at the higher frequencies.

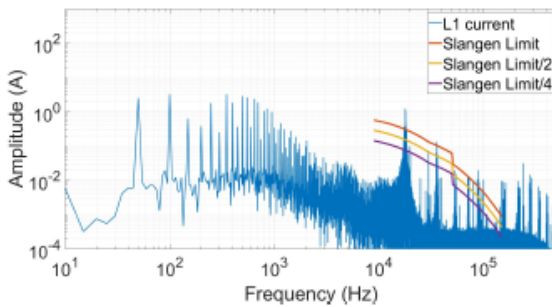


Fig. 14: EVSE 3: Frequency spectrum while charging at 450 V, 0 A.

Figure 15 shows the SHC in the 18 kHz bin with a bin size of 2 kHz of EVSE 3 relative to the current and voltage. The distortion is highest while charging with 0 A below 800 V. From 50 A, the SHC increases with current. The distortion is lowest between 850 and 900 V.

Figure 16 increases the DC current resolution in the 0 - 50 A range. The SHC is raised significantly if the EVSE is charging at 0 A. If a charging plaza with only this type of EVSE must temporarily interrupt charging due to grid congestion, the SHD emitted to the grid could become substantial.

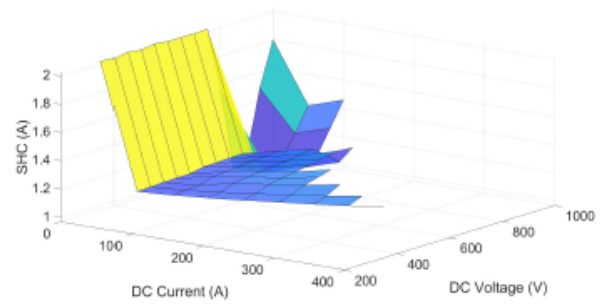


Fig. 15: EVSE 3: SHC in the 18 kHz bin.

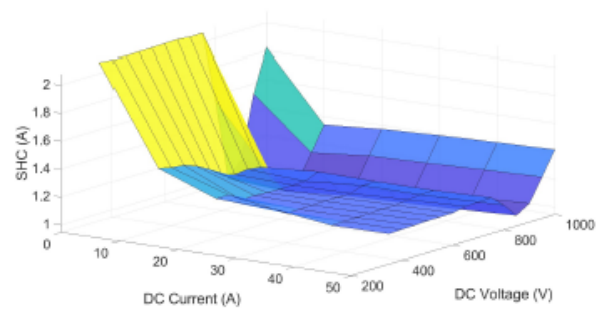


Fig. 16: EVSE 3: SHC in the 18 kHz with higher DC current resolution.

D. Comparisson

The spectra of EVSE 2 and 3 are similar, which was unexpected. They both contain a single dominant, narrowband frequency component and its harmonics. EVSE 1's spectrum differs from the others as anticipated, since it mainly contains wideband distortion.

The wideband high-frequency distortions are present in EVSE 1 and 3 and occur at similar frequencies. It seems they are also present in EVSE 2's spectrum, although it is hard to see due to the raised noise floor. It is assumed they are caused by the measurement setup because they occur in all presented spectra at similar magnitudes, while other parts of the spectra vary from each other. The same goes for the 240 kHz distortion. Since the 100 kHz distortion is assumed to be caused by the measurement setup, EVSE 1 complies with the installation level Slangen limit, and nearly with the 25% limit. EVSE being the closest to compliance is unexpected as it is the highest power EVSE. Since EVSE 1 nearly complies with the 25% limit, they seem realistically achievable for EVSE manufacturers.

The characteristics of the EVSEs' SHC per operating condition vary significantly, as expected. EVSE 1's SHC seems to increase with power, EVSE 2's SHC appears to lack any correlation, and EVSE 3's SHC seems highest when supplying 0 A.

V. CONCLUSION

Three EVSEs have been measured to characterize their SHD at the grid side. The knowledge can contribute to the division of standards for the SH frequency range. Aside from 2 EVSEs showing similar spectra, the data showed significant differences between EVSEs in spectral behavior and SHC characteristics for various operating conditions. The measurements also indicated that distortion can vary in frequency and amplitude over time. The EVSEs have also been tested against the Slangen limits assuming the grid emulators had a relatively high impedance. Two EVSEs exceeded the installation limit. However, the EVSE with the highest power almost remained within the individual device limit. Hence, the individual device limit seems feasible.

VI. RECOMMENDATIONS

This section lists the author's future work recommendations.

- It is recommended to research under what conditions the time-dependent emissions of EVSEs occur, and if these can be removed from a standardized compliance test.
- More research is required on how the RBW influences the repeatability of compliance test results in the SH range, and if band overlap or time domain EMI receivers are (feasible) solutions.

REFERENCES

- [1] T. M. H. Slangen et al. "The Harmonic and Supraharmonic Emission of Battery Electric Vehicles in The Netherlands". In: *2020 International Conference on Smart Energy Systems and Technologies (SEST)*. 2020, pp. 1–6. DOI: 10.1109/SEST48500.2020.9203533.
- [2] Frank Leferink. "Conducted Interference, Challenges and Interference Cases". In: *IEEE Electromagnetic Compatibility Magazine* 4.1 (2015), pp. 78–85. DOI: 10.1109/MEMC.2015.7098517.
- [3] CENELEC SDC 205A Study Report on Electromagnetic Interference between Electrical Equipment/Systems in the Frequency Range below 150 kHz. Tech. rep. SC205A/Sec0329/DC. CENELEC, March 2013.
- [4] Sarah K. Rönnberg et al. "On waveform distortion in the frequency range of 2kHz–150kHz—Review and research challenges". In: *Electric Power Systems Research* 150 (2017), pp. 1–10. ISSN: 0378-7796. DOI: <https://doi.org/10.1016/j.epsr.2017.04.032>. URL: <https://www.sciencedirect.com/science/article/pii/S0378779617301864>.
- [5] IEC-61000-4-7:2002/A1:2009. *Electromagnetic compatibility (EMC) - Part 4-7: Testing and measurement techniques - General guide on harmonics and interharmonics measurements and instrumentation for power supply systems and equipment connected thereto*. 2009.
- [6] T. M. H. Slangen et al. *Standards for 2-150 kHz*. Tech. rep. TEPQEV, 2023. URL: <https://elaad.nl/wp-content/uploads/downloads/TEPQEV-Whitepaper-standardization-20230228.pdf>.
- [7] Tim Slangen et al. "Variations in Supraharmonic Emission (2-150 kHz) of an EV Fast Charging Station under Different Supply- and Operating Conditions". English. In: *2023 IEEE Power & Energy Society General Meeting (PESGM)*. 2023 IEEE Power & Energy Society General Meeting (GM) ; Conference date: 16-07-2023 Through 20-07-2023. United States: Institute of Electrical and Electronics Engineers, Sept. 2023, pp. 1–5. DOI: 10.1109/PESGM52003.2023.10252962.
- [8] Jon González-Ramos et al. "Emissions generated by electric vehicles in the 9-500 kHz band: Characterization, propagation, and interaction". In: *Electric Power Systems Research* 231 (2024), p. 110289. ISSN: 0378-7796. DOI: <https://doi.org/10.1016/j.epsr.2024.110289>. URL: <https://www.sciencedirect.com/science/article/pii/S0378779624001779>.

APPENDIX A

GRID IMPEDANCE MEASUREMENT

Figure 17 shows the measured grid impedance at the EVSE's point of connection, which means the impedance of the cables is included. It shows an inductive impedance. 4 AC grid emulator (ACE) modules were used for the EVSE 1 measurements, as 3 ACE modules could not provide enough power. During the measurements of EVSE 2 and 3, only 3 ACE modules were used.

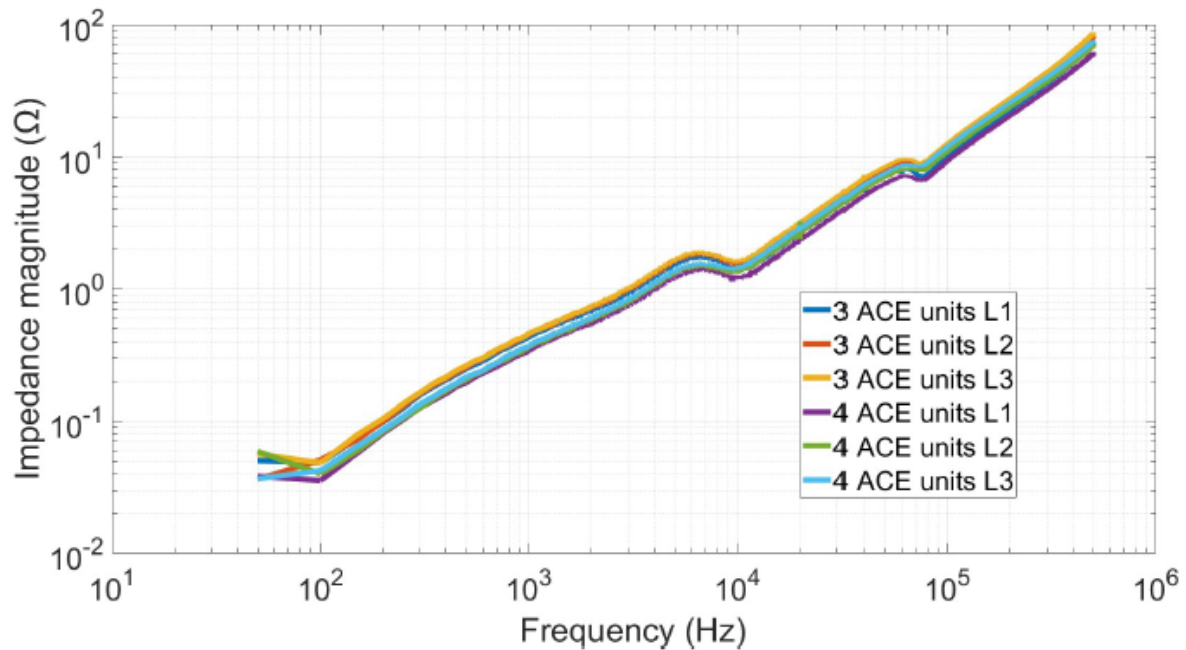


Fig. 17: Measured impedance of the grid emulator with no load connected.

Appendix B

Vehicle

2023

(vehicle name) Charging Test

Measuring and analysing the (smart) charging process of electric vehicles



Thijs van Wijk | Remi Keijsers

ElaadNL

Test dates: ____ till ____

Report date: ____

datum; ElaadNL

Management summary

ElaadNL is a foundation set up and financed by the Dutch electricity grid operators. Our goal is to smoothly integrate electric vehicles and charging infrastructure in the electricity grids. One of the ways we do this is by testing electric vehicles at the ElaadNL Test Lab. In this lab we test electric vehicles on their smart charging behaviour, on their interoperability with the Dutch charging infrastructure and on their power quality (PQ) emission and immunity. Also, in respect to safety we check if the cable is always locked during charging and if no voltages and currents are present on the charging cable where and when there should not be.

Recently we tested the _____ on these topics. Find below an overview and the main conclusions of these tests. The colours in the table below show if the vehicle passed all tests or if there were some remarks and/or some room for improvement.

Test	General/Safety	Interoperability	Smart charging	PQ emission	PQ immunity
General result					

Legenda	
Good	
With remarks	
Improvements needed	

The main conclusions are;

The other tests went without issues. The vehicle is not affected by ...

Final words

The remarks found are important because they can cause charging issues during smart charging which is commonly used throughout The Netherlands and when the voltage in the grid is not optimal, but still within limits.

We recommend you to have your technicians read the reports in detail, to clearly understand the tests we perform, the results and our findings. The document starts with an explanation of the tests, followed by the test forms in the annexes and ends with several screenshots of the measurements. If there are any questions about the tests or the results, feel free to contact us at testlab@elaad.nl. Also, by request we can make the measurement data available for download.

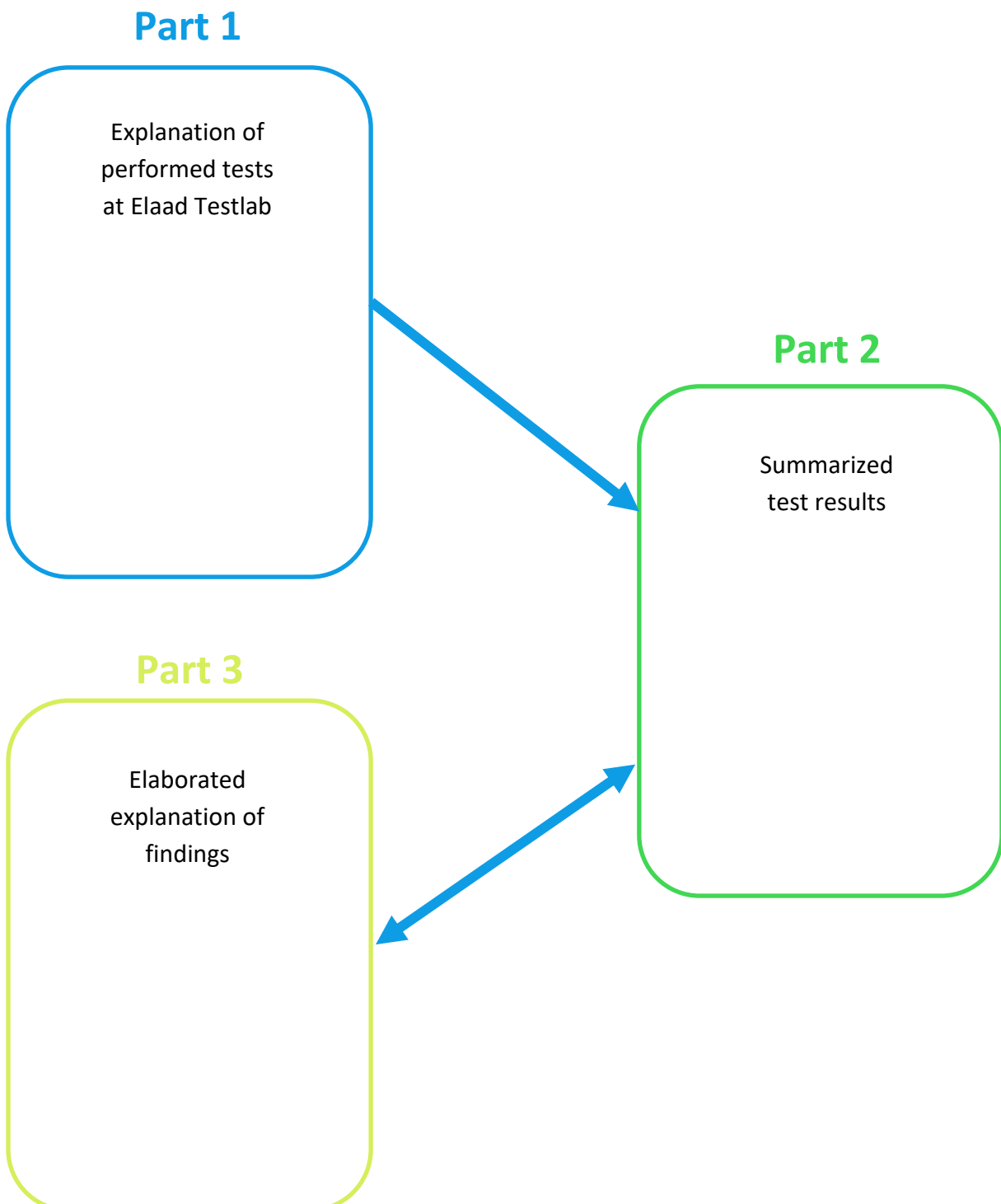
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Document explanation

Please be aware this document is mainly built up of 3 parts. The first part is explaining all the different tests that are performed at the Elaad Testlab. The second part will be all the testcase results summed up divided by each category. The last part will explain in greater detailed about all findings during the testing. This part consists of clear screenshots with elaborate explanations about the findings. We'd recommend to first read part 1 to get a better understanding of the tests performed at the Elaad Testlab. We'd recommend viewing part 2 and 3 next to each other to get the best overview of all the test results.



Test overview

This document describes the interoperability, smart charging and power quality tests performed at the ElaadNL test site. In general, these tests are performed to identify any issues concerning the integration of the Electric Vehicle (EV) with the charging infrastructure and the electricity grid. Per subject, see overview below, several different tests are performed of which each has a specific purpose. These are explained in detail in the next chapters.

- General and safety tests; testing the general charging behaviour of the EV, its abilities and the safety during charging to improve customer experience and to match the other tests with the abilities of the EV.
- Interoperability tests; testing the EV's ability to charge flawlessly on the different charging stations during regular and smart charging.
- Smart charging laboratory tests; tests and measurements on how the EV reacts to changes in the PWM signal and different smart charging scenarios, like fluctuating charging speeds, intermittent and delayed charging.
- Power Quality emission tests; tests and measurements of the influence the charging EV has on the power quality in the electricity grid.
- Power Quality immunity tests; test and measure the influence the quality of the grid itself has on the charging behaviour of the electric vehicle.

The following tests are advised, but the OEM is free to perform other tests as well. It is advisable to test on all the available stations, because of possible differences in mode 3 implementations and timing settings.

1. General EV behaviour and safety tests

During the General and safety test the EV will be tested on its general charging behaviour and if the safety during charging can be improved. The tests will be performed using a charging station at the ElaadNL testing ground and, when supplied with the vehicle, a mode 2 charging cable.

- Vehicle behaviour test:
 - o What triggers the vehicle to stop/start charging?
 - o On how many phases can the vehicle charge and on what maximum current?
- Safety during regular usage;
 - o Plug removal
 - o Charging without protective earth conductor
- Vehicle response to absence of or low voltage and safety
 - o Total absence of voltage
 - o Phase independent absence of voltage at start transaction
 - o Phase independent absence of voltage during a transaction
 - o Low voltage at phase 2 & 3 at start transaction

These tests are further explained in paragraph 1.1 till 1.3. A test form to be used during these tests can be found in annex A.

1.1. Vehicle behaviour

This procedure is done once at one single charging station, to find out which events will trigger the vehicle to start or stop charging (lock/unlock, close/open door), if a certain plug-in order is expected by the vehicle, and if the vehicle can charge on single, double and triple-phase and at what

maximum charging speed. The acquired information will be used to advise on vehicle behaviour improvements to help customer to charge their vehicles in an intuitive and safe manner and is used to match the other tests with the capabilities of the vehicle. Also, the information is used to list how much current the vehicle is able to take in during charging on 1 phase and 3 phases. Regarding 1 phase charging; as a general rule in The Netherlands (Dutch *Netcode elektriciteit* of 14-12-2021) any device taking in more than 5,75 kVA (~25A) should be connected via three phases to prevent voltage unbalance. In Germany a similar but more strict rule applies, the VDE-AR 4100 standard, which states that the maximum connection power for single-phase loads is limited to ≤ 4.6 kVA (~20A) when connecting to the low-voltage grid of power utilities.

These limits might result in charging stations being limited in power for 1 phase vehicles to 25A or 20A. Because this further limits the total charging speed for the customer, and to prevent voltage unbalance in the grid, ElaadNL advises to use 3 phase on-board chargers.

Estimated test time: 30 minutes

1.2. Safety during regular usage

During these tests the focus is on both personal and technical safety.

- A check is performed to see if the cable is always locked at the vehicle during charging to prevent possible unsafe situations, possible wearing of the conducting pins due to an electric arc and to prevent cable theft. ElaadNL strongly recommends locking the cable because of the above reasons. The test is performed in accordance with ISO 17409; *Electrically propelled road vehicles — Conductive power transfer — Safety requirements*. The timing and current in the screenshot below from ISO17409 are used as boundaries.

8.4 Interlock function for the vehicle coupler

When equipped with a vehicle inlet that is not suitable for making and breaking an electrical circuit under load, the vehicle shall provide an appropriate interlock function.

- When the proximity detection circuit is used for such an interlock function, the vehicle shall stop power transfer operation and reduce the current through the vehicle coupler to less than or equal 1 A within 100 ms after actuation of the switch in the proximity detection circuit.
- When locking is used for such an interlock function, the vehicle shall inhibit release of the vehicle connector while the vehicle current exceeds 1 A.

Figure 1: ISO17409 2020-02 on interlock function

- Regarding cable theft an additional test is performed to check if the cable remains locked at the vehicle side during a power outage.
- Lastly, regarding personal safety a test is performed on a regular charging station and, if available, on a mode 2 charger supplied with the vehicle, without a protective earth conductor being present. This test is performed because charging without a protective earth conductor is not allowed and can lead to the build-up of static electricity on the vehicle's chassis. This is also stated in the 61851-1; for mode2, 3 and 4 chargers it is mandatory to do *Continuous continuity checking of the protective conductor*.

Estimated test time: 1 hour

1.3. Vehicle response to absence of or low voltage and safety

During the following test the EV its response to absence or low voltages on all or individual phases is tested. Apart from learning about the EVs ability to (restart) charging when phase voltages change, checks are performed to see if the EV puts back voltages on non-powered phases. This can, for instance, be the case if the vehicle is able to switch multiple convertors to one phase when only this phase is available and does this in such a way that the voltage of the one phase is transferred via the linked convertors to the other, non-powered phase. This is what we call phase-linking. An EV is not allowed to put any voltages back on the cable to the charging station (if it is not a V2G charging session). These unexpected voltages can lead to unsafe situations, like a short circuit when the voltage on the phase is restored or an electric shock to people working on the installation.

1.3.1 Total absence of voltage

During the following tests the response of the vehicle on a total absence of voltage on the phases is observed. It is possible for the charging station to be powered but not able to provide power for charging, e.g. contactor failure.

Test	Required/advised behaviour
a.	Station powered but unable to provide power momentarily during charging
b.	Station powered but unable to provide power at the start of a transaction

1.3.2 Phase independent absence of voltage at start transaction (only on multi-phase vehicles)

It is possible that at the start of a transaction one or more phases do not have any voltage. It is tested how the vehicle responds when the voltage is absent on one or more phases and how the vehicle behaves when voltage is brought back on the missing phases. Special attention is given on possible phase-linking when charging a multiple phase vehicle on 1 phase.

1.3.3 Phase independent absence of voltage during a transaction (only on multi-phase vehicles)

It is possible that during a transaction one or more phases lose their voltage. It is tested how the vehicle responds when the voltage is absent on one or more phases and how the vehicle behaves when voltage is brought back on the missing phases. Special attention is given on possible phase-linking when charging a multiple phase vehicle on 1 phase.

1.3.4 Low voltage at L2 and L3 at start transaction (only on multi-phase vehicles)

This test is performed to monitor the behaviour of a vehicle when there is an extremely low voltage on L2 and L3. Voltage is brought back to 150V, 100V and 50V. Vehicle is expected to

Vehicle

stop charging at these low voltage levels. Special attention is given on possible phase-linking when supplying such a low voltage on L2 and L3.

Estimated test time: 1,5 hours

2. Interoperability tests

During the interoperability test the EV will be tested on its specific, per charging station, charging behaviour. These tests are further explained in paragraph 2.1. The tests will be performed using the charging stations at the ElaadNL testing ground. An overview of the charging stations and their position in the testing ground can be found in paragraph 2.2 and 2.3.

A test form to be used during these tests can be found in annex B.

2.1. Mode 3 interoperability testing procedures

- (Smart) charging behaviour:
 - o Regular charging test: Scope of this test is to find mode 3 communication issues (for example; limits of the mode 3 voltages of EV and charging station are not aligned)
 - o Smart charging test: this test includes changing the charging current levels offered by the stations and interrupting charging for 5 minutes. Scope of this test is to find possible reaction time related issues on changes to the signalled current level and to see if the vehicle is able to “wake-up” after a charging interrupt.
- Station behaviour on vehicle charging stop: Scope of this test is to see how vehicle and charging station react to each other when the vehicle stops charging.

2.1.1 (Smart) charging behaviour

Regular charging test

During the test on the different station types, the following routine procedure is used to see how the vehicle responds in combination with the specific station:

- In between the charging cable and station connector a break-out connector box is placed which allows to measure the CP (Communication Pilot) signal.
- An oscilloscope measurement tool is connected to the CP cable in the break-out connector box to measure the mode 3 communication and the behaviour of the PWM signal.
- A transaction is started using a regular RFID charging card.
- A check is performed on the mode 3 steps (12, 9, 6 volt and PWM start)

Estimated test time: 4 hours.

Smart charging test

If the station is able to smart charge, a charging profile is sent to the station. The station will change its PWM signal to match the different current levels defined in the charging profile. The following maximum allowed currents will apply during the test. These maximum currents might be lower depending on the capabilities of the station.

Start time (s)	0	30	60	90	120	150	180	480
Current (A)	25	6	25	0	6	25	0	16

- During the first minute, the current will fluctuate between 25 and 6 Amps. This is done to test the vehicle’s response to a “smart” charging session.

Vehicle

- After 90 seconds, the vehicle is tested on how it responds when the charging session is paused for a few seconds and how it responds when the vehicle starts charging again after the interruption. It is worth to note that not all charging stations pause a charging session the same way. Some stations pause by going to state B1 (9V, 100%PWM), others to state F (-12V, 0%PWM), and even others first go to state B1 and after a few seconds to state F.
- After 3 minutes, the session will be interrupted for 5 minutes in total. The purpose of this test is to check if the vehicle will not fall asleep after lowering the current to 0 for a longer period of time. The vehicle will pass the test when it starts charging again after a 5-minute interruption.

The following profile will be executed on stations which have different pausing mechanisms discussed in the text of the previous profile. This time the pausing period will be 20 minutes. The current will be increased afterwards from 6 to 8 to 16A, to test if the vehicle does need a certain minimal allowed current to start charging again.

Start time (s)	0	1200	1260	1320
Current (A)	0	6	8	16

Estimated test time: 8 hours.

2.1.2 Station behaviour on vehicle charging stop

The response of the charging station when the vehicle stops charging is also tested. This can be tested in 3 different ways. These tests will check if the vehicle signals the station it stopped charging (by opening s2) and if the station in this case will not send multiple state E or F statuses to “wake-up” the car.

- A first method is reaching a desired battery level. This test can be done by either reaching 100% or any other predefined max charge level.
- A second (optional) method is delayed charging. This must be supported by the vehicle. This basically means the vehicle delays the charging according to a predefined setting. When the vehicle is ready to charge again it should close its switch and the charging station should react.
- A third (optional) method is stopping at vehicle side. For example, some vehicles are capable of stopping/pausing the charging session by unlocking the vehicle.

Estimated test time: 4 hours.

2.1.3 Vehicle behaviour on different protocols

Generally, all tests described in this document will be performed using the 61851 “mode 3” or “mode 2” protocol. A short check will be done to see if the vehicle will also support different protocols like the DIN 70121 and 15118 protocol. Even if not yet completely implemented, a check of the current status of the implementation will be done.

Estimated test time: 2 hours.

2.2. ISO 15118 interoperability testing procedure

The Elaad Testlab has several chargers which are able to communicate via the ISO15118-2 protocol. Most of them are DC chargers, which are able to communicate via DIN 70121 or ISO15118. During the interoperability test a charging test on these stations will be performed. Also, Elaad has an AC 15118-2 charging station from Trialog, which it can use for testing 15118-2 interoperability at an AC charging station.

2.2.1 DC 15118-2 interoperability test

The vehicle will be tested on the different DC chargers at the Elaad Testlab. It will be tested if the vehicle charges at the DC charger and, if the logging of the chargers allows, the communication method used will be logged. Also, if the station allows smart charging, a charging profile will be sent comparable to the one used during the AC smart charging interoperability test.

2.2.2 AC 15118-2 interoperability test

The AC 15118 charging test will be performed using the Trialog 15118-2 AC charging station at Elaad. This station can be directly controlled by Elaad and via this way several tests can be performed. The following items are being checked at different security levels (if supported).

- General communication setup ok?
- Unsecure communication;
 - o smart charging
- Secure communication (TLS only)
 - o Smart charging
 - o Plug & Charge + Smart Charging
 - o Plug & Charge + Certificate install.
- Secure communication (TCP & TLS)
 - o Smart charging
 - o Plug & Charge + Smart Charging
 - o Plug & Charge + Certificate install.

Estimated test time: 2 hours.

2.3. Station overview

DC charging stations

Manufacturer	Type	Sockets	Protocols	kW/socket	Voltage	Country
Ambibox	Ambicharge	1 (CCS2)	70121, 15118	11kW	120-500V	Germany
Autel	MaxiCharger DC Fast	2 (CCS2)	70121, 15118	240 (80 for now)	150-950V	China
Autel	MaxiCharger DC Wall Mount	2 (CCS2)	70121, 15118	40 (Max 47)	150-1000V	China
Delta	EVDE25E	1 (CCS2)	?	25	50-500V	Netherlands
eNovates	DC V2G	1 (Chademo)	Chademo	11		Belgium
Heliox	40kW mobile	2 (CCS2)	70121, 15118	40	100-1000V	Netherlands
Heliox	Rapid 150	2 (CCS2)	70121, 15118	150 (40 for now)	150-1000V	Netherlands
Heliox	Flex 180	1 (CCS2)	70121, 15118	180 (40 for now)	150-1000V	Netherlands
Ingeteam	Ingerev rapid 180	2 (CCS2)	70121	180 (40 for now)	150-1000V	Spain
Kostal	DC 11kW	1 (CCS2)	70121, 15118	11	Max 450V	Germany
Nio	Nio Power	1 (CCS2)	70121	180	Max 1000V	China
Vestel	EVC06	2 (CCS2)	70121	60kW(40 for now)	200-920V	Turkey
Vestel	EVC03	2 (CCS2)	70121	180	200 – 920 V	Turkey

(Semi) Public AC charging stations

Manufacturer	Type	Sockets	Protocols	kW/socket	Country
Alfen	EVE Double Pro-Line	2	Mode3	22 (Max)	Netherlands
Alfen	Tube 2.0 (NG910)	1	Mode3	11	Netherlands
Alfen	Twin XL	2	Mode3	11 (Max)	Netherlands
Alfen	Twin 4XL	2	Mode3	27,6 (Max)	Netherlands
Compleo	Duo	2	Mode3	22 (Max)	Germany
Connected Kerb	Chameleon	2	Mode 3	22	UK
Ecotap	Yeris wide 64A	2	Mode3	22 (Max)	Netherlands
eNovates	Pro22	2	Mode3	11	Belgium
eNovates	Pro11	1	Mode3	11	Belgium
EV-Box	Business line	1	Mode3	17,2	Netherlands
EV-Box	Elvi	1	Mode3	22	Netherlands
EV-Box	ML zuil 64A	2	Mode3	22 (Max)	Netherlands
GE	Durastation (LMS)	2	Mode3	11 (Max)	Netherlands
Reewoud	Chargepoint P6 V2G (LMS)	2	Mode3	11 (Max)	Netherlands
RWE	Innogy	2	Mode3	11 (Max)	Germany
Smappee	EV Base	2	Mode3	22 (Max)	Belgium
Trialog	AC15118	1	Mode3/15118-2	11 (Max)	France
Vestel	EVC05	2	Mode 3	22	Turkey

AC Wallboxes

Manufacturer	Type	Sockets	Protocols	kW/socket	Country
Alfen	EVE single Pro-line	1	Mode3	22	Netherlands
Autel	MaxiCharger AC Wallbox	1	Mode3	22	China
Connected Kerb	Scarab	1	Mode3	22	UK
CTEK	Chargestorm Connected 2	2	Mode3	22 (Max)	Sweden
Easee	Home	1	Mode3	22	Norway
EO Charging	Genius 2	1	Mode3	22	UK
Etrrel	Inch home	1	Mode3	22	Slovenia
Liteon	EU 32A intelligent charger	1	Mode3	22	Taiwan
Schneider Electric	EVLink smart wallbox	1	Mode3	22	France
Schneider Electric	EVLink Pro	1	Mode3	7,4	France
Smappee	EV Wall	1	Mode3	22	Netherlands
Vestel	EVC04	1	Mode3	22	Turkey
Wallbox	Copper SB	1	Mode3	22	Spain
Wallbox	Commander 2	1	Mode3	22	Spain
Zaptec	Go	1	Mode3	22	Norway
Zaptec	Pro	1	Mode3	22	Norway

3. Smart charging laboratory tests

Smart charging tests are both performed during the interoperability tests in the testing ground and during the Smart Charging laboratory tests in the Test lab. The tests on the proving ground give a more general picture, while in the test lab the charging process can be monitored more closely and detailed and more scenarios can be tested.

The following tests are meant to measure the time needed by the EV to adapt to PWM changes, to see what charging speeds are accepted and to perform a check on the ability of the EV to handle different smart charging scenarios. These tests are performed using mode 3 communication as described in the IEC 61851-1 (2019).

The IEC 61851-1 standard states the EV shall adjust its maximum current drawn to be equal or below the maximum current indicated by the PWM duty cycle within 5 seconds after the station signals the new maximum via the PWM signal. Also, it states that the vehicle needs to stop drawing current within 3 seconds after the station stops the PWM signal and afterwards the vehicle has 3 seconds to open its S2 switch.

A test form to be used during the test can be found in annex C.

3.1. Fast changing currents tests

These tests measure the response of the vehicle to smart charging profiles of different kind.

- a. **Fluctuating charging profile:** testing the behaviour of the vehicle to changes of different size in the allowed charging current and the ability to charge on different charge current levels.

Start time (s)	0	60	120	180	240	300	360	420	480
Current (A)	0	6	9	12	16	20	24	28	32
Start time (s)	540	600	660	720	780	840	900	960	1020
Current (A)	28	24	20	16	12	9	6	32	6
Start time (s)	1080	1140	1200	1260	1320	1380	1440	1500	
Current (A)	32	6	0	32	0	32	0	16	

- b. Fluctuating with no charge in between:** This test is performed to check how the EV reacts to a short *no charge* in between changes in the PWM signal. This behaviour is tested because some types of chargers smart charge this way. A *no charge* time of five seconds has been chosen here because this confirms to the five seconds an EV must change its charging current according to the IEC 61851-1 standard.

Start time (s)	0	55	60	115	120	175	180	235	240
Current (A)	6	0	10	0	16	0	20	0	32

Start time (s)	295	300	355	360	415	420	475	480
Current (A)	0	10	0	6	0	32	0	16

Estimated test time: 1 hour.

3.2. Intermittent charging test

This test measures the response of the vehicle to intermittent charging, as can be used as load balancing option by charging stations. To perform this test a charging profile containing frequent pausing periods is send to the stations, to make the vehicle charge and pause charging. The interval between the changes is set to 60 seconds. A single test is performed that switches between 6 and 0 Amps for 25 times. If the vehicle is not able to charge at 6A, a higher value will be selected.

Intermittent charging 0-6 Ampères

Start time (s)	0	60	120	180	240	300	360	420	480	540	600
Current (A)	6	0	6	0	6	0	6	0	6	0	6

Start time (s)	660	720	780	840	900	960	1020	1080	1140	1200	1260
Current (A)	0	6	0	6	0	6	0	6	0	6	0

Start time (s)	1320	1380	1440	1500	1560	1620	1680	1740	1800	1860	1920
Current (A)	6	0	6	0	6	0	6	0	6	0	6

Start time (s)	1980	2040	2100	2160	2220	2280	2340	2400	2460	2520	2580
Current (A)	0	6	0	6	0	6	0	6	0	6	0

Start time (s)	2640	2700	2760	2820	2880	2940	3000	3060
Current (A)	6	0	6	0	6	0	8	0

Estimated test time: 1,5 hours.

3.3. Delayed & paused charging

This test checks if an EV is able to restart charging when being delayed and paused for longer time periods. Two types of tests are performed; delayed charging, during which the charging is delayed from the start of the transaction, and paused charging; during which the vehicle is allowed to charge for a moment and then paused. Two time periods are used for these tests.

First, a 20-minute delay and 20-minute pause test is performed in the lab. 20 minutes has been chosen to test if the vehicle can cope with a 15-minute delay or pause as is commonly used during intermittent charging.

a. Delayed charging 20 minutes

Start time (s)	0	1200
Current (A)	0	16

b. Paused charging 20 minutes

Start time (s)	0	60	1260
Current (A)	16	0	16

Secondly, if the vehicle performs well during the 20-minute test a 6-hour delayed and paused test is performed to test if the vehicle can cope with longer delays and pauses, like being used for delaying or pausing the charging session to after the evening electricity usage peak. To perform these tests an overnight test is set up using a charging station that uses state B1 when no energy is available. A charging profile is sent to the station that first delays the charging session for 6 hours, shortly starts charging at 6A for 5 minutes, and pauses the charging session again for 6 hours, after which charging is allowed again for 5 minutes.

c. 6 hours delayed and paused charging

Start time (s)	0	21600	21900	43500	43800
Current (A)	0	6	0	6	0

It is possible a vehicle contains systems which block the vehicle from going to sleep mode. These systems should be turned off.

If the car doesn't wake up correctly new tests will be done (if time allows) to pinpoint the maximum allowed delay/pause times and the possible influence of the turned off systems.

d. 3 hours low charging speed

Start time (s)	0	10800
Current (A)	6	16

It is possible a vehicle does not start drawing more current when it is allowed to do so. Therefore a test will be done where the vehicle is allowed to charge for 3 hours on 6A, after which it is allowed to charge for 16A.

If the car doesn't wake up correctly new tests will be done (if time allows) to pinpoint the maximum allowed delay/pause times and the possible influence of the turned off systems.

Estimated test time: 9,5 hours.

3.4. PWM signal test

This test checks the response of the vehicle to a certain set of critical outer limit PWM signals. To perform this test the PWM signal from the station is directly controlled to send the outer PWM limits.

PWM	Required behaviour
0 < 3	Charging not allowed
3 ≤ Din ≤ 7	Digital communication
7 < 8	Charging not allowed
8 ≤ Din < 10	6A
10 ≤ Din ≤ 85	Din * 0,6A

Estimated test time: 5 minutes.

3.5. Accessory circuit power tests

This test is performed to check if the vehicle does not exceed current limit set by the charging station when the accessory circuits are activated. Tested at the minimal current limit (6A) and at a limit corresponding to the max nominal current of the vehicle. Possible accessory circuits are (if available); climate control, heated seats/steering wheel and retractable roof/powered windows (with the radio installation as a bonus).

Charging speed	Any noticeable changes in current?
6	
Max	

Estimated test time: 10 minutes.

4. Power Quality emission tests

To gather information about the influence charging has on the electricity grid, the EV will be charged while the charger currents, voltages and the mode3 signal are being measured in detail using a Dewetron DEWE-800 measuring computer in combination with LEM IT65-S current transducers, see figure 3. The Dewetron has a 1 mega-sample sampling rate with 16-bit resolution of all currents and voltages in the 3-phase grid, which makes it able to measure in great detail, and will store all measured data for analyses. The current transducers have a 600 kHz bandwidth and are made for measuring with very high accuracy and linearity.



Figure 2: Dewetron DEWE-800 and LEM IT65-S

The measurements will be performed on our local grid and on our Chroma 61830 grid emulator, see figure 4. During the power quality emission tests the grid emulator is used to create a stable, perfect 230V 50 Hz grid without any harmonic distortions. This way it can be ruled out that any measured emissions or anomalies are caused by other devices connected to the grid. Only the Supraharmonic emission measurement is both performed on the grid emulator as well as on the grid, to learn more about the impact of these type of distortions on a real grid.



Figure 3: chroma 61830

The following Power Quality effects will be measured during the tests. A test form to be used during the test can be found in annex D.

4.1. THD and the first 40 harmonics

Current and voltage in an AC grid should ideally perform a perfect sine wave. Not all devices do take in current as a perfect sine wave because of their internal components. Because current and voltage influence each other, a non-perfect current sine wave will also cause the voltage to become less perfect. And when the voltage is not a perfect sinus anymore, the same effect will be present on the current taken in by the other connected devices. In the figure below you will find an example of a perfect voltage sinus (left) and one of a distorted sinus (right)

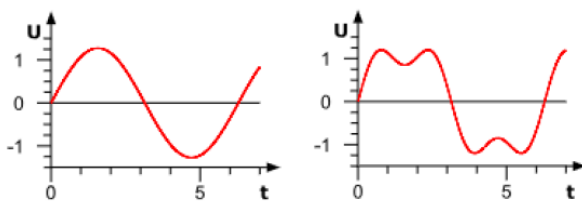


Figure 4: perfect and distorted voltage sinus

If these currents are of a frequency that is an integer multiplication of the fundamental frequency of the grid (50Hz), it is called a harmonic current. So, the 2nd harmonic is at 100 Hz, the 3rd at 150 Hz, etc. During this test the amplitude of the different harmonic currents that are present during different charging speeds will be measured. According to the IEC-61000-3-2/12 standard the following maxima (next page) are allowed per harmonic. There are different levels for currents under

datum; [ElaadNL](#)

16A, and currents between 16 and 75A. The vehicles are hereby tested as equipment other than balanced three-phase equipment. This goes for three phases vehicles too, as 3 phase converters are non-balanced devices.

Under 16A current input per phase

As defined in the IEC-61000-3-2, the maximum height of the different harmonic currents (up to the fortieth), is shown in the table below.

Even - Harmonic	Max current	Odd-Harmonic	Max current
2	1,08	3	2,3
4	0,43	5	1,14
6	0,3	7	0,77
8	0,23	9	0,4
10	0,18	11	0,33
12	0,15	13	0,21
14	0,13	15	0,15
16	0,12	17	0,13
18	0,1	19	0,12
20	0,09	21	0,11
22	0,08	23	0,1
24	0,08	25	0,09
26	0,07	27	0,08
28	0,07	29	0,08
30	0,06	31	0,07
32	0,06	33	0,07
34	0,05	35	0,06
36	0,05	37	0,06
38	0,05	39	0,06
40	0,05		

In between 16A and 75A current input per phase

As defined in the IEC-61000-3-12, the maximum height of the different harmonic currents (up to the thirteenth) and the Total Harmonic Current (THD), is shown in the table below.

Harmonic	Max percentage
2	8%
3	21,6%
4	4%
5	10,7%
6	2,67%
7	7,2%
8	2%
9	3,8%
10	1,6%
11	3,1%
12	1,33%
13	2%
THD	23%

4.2. Supraharmonic distortions

Apart from the Total Harmonic Distortion (THD) and the first 40 harmonics, which allowed maxima are defined in the IEC-61000-3-2/12 standard, the *supraharmonic* distortions are measured. Supraharmonic distortions range in the 2 till 150 kHz frequency band. This frequency spectrum is not regulated in any standards yet. Just like (inter)harmonic distortions, supraharmonic distortions can cause an increase of temperature in grid assets and can influence the grid voltage, which might lead to failure of grid assets, charging stations, (consumer) electronics and electric vehicles themselves. It is therefore good practice to keep the supraharmonics as low as possible as well. As there are no standards for the Supraharmonic range yet, there are no limits available. During the analysis of the Supraharmonics we therefore will not check for limit breaches, but we will register the frequency, height and possible shifting behaviour of the distortion. This information will be shared with the vehicle manufacturer and will anonymously be used in further research on Supraharmonic distortions and as input for the standardisation activities.

4.3. Supraharmonic interaction (requires two identical vehicles)

When 2 or more EVs with a similar charging system are charged at the same time and nearby each other, supraharmonic interaction via the grid can occur. If there is the opportunity to run the test with two of the electric vehicles under test, we can test if the vehicles are immune to each other's Supraharmonic emission (if applicable) and if the emissions change. A known effect for instance is frequency beating; This phenomenon occurs when at least 2 (nearly identical) EVs are charged together, of which the Supraharmonic emission frequencies lay close to each other. This may result in a distortion that will fluctuate in strength following the small difference in emitted frequencies by the vehicles. This can result in disfunction of protection circuits and/or additional heating and possible audible noise nearby the EV or in switching cabinets.

4.4. Reactive currents/power factor

Reactive currents are currents that are transported, but are out of phase when compared to the voltage. This means they cannot be used to generate power or, in this case, to charge the vehicle. The ratio of reactive current is shown via the power factor which shows the ratio between the real currents and the reactive currents. The lesser the reactive currents, the higher the Power factor. The Dutch DSOs demand that the power factor of an installation must be at least 0.85 (see article 2.27 from the Dutch "Netcode Elektriciteit") at voltage up to 50kV at a scale from 0 to 1. Typically, the higher the power factor, the better. Both because the grid capacity will be used more efficiently and the vehicle will be charged faster.

4.5. Inrush currents/transients

At the start of a charging session a AC/DC converter might take in a short, but high amount of current. For instance, to charge up its internal capacitors. The measuring data is analysed to find if inrush currents or transients are present and if it is of a height and length that might cause issues with the MCBs used for overcurrent protection in the charging stations. Typically, an inrush current of 3 times the nominal current is considered to be workable, but the lower the better.

5. Power Quality immunity tests

Apart from the influence the electric vehicle has on the power quality of the grid, the quality of the grid itself can also influence the charging behaviour of the electric vehicle. For instance, a sudden drop in the voltage might make the vehicle stop charging. At the ElaadNL testlab this can be tested by using the Chroma 61830 grid emulator. This device is able to emulate a grid of predefined specifications and gives the ability to change the voltage height, frequency and quality for all three phases separately during the tests. The test described below are performed while the vehicle charges on its minimum and maximum charging current and at 1 and 3 phase charging (depending on the capabilities of the vehicle). A test form to be used during the test can be found in annex E.

5.1. Grid voltage tests

The grid is not always completely stable. It is possible the voltage sometimes is lower or higher than desired. From a customer perspective it would be preferable that the vehicle will not abort the charging and will continue when the voltage is within the allowed 10-minute average voltage deviations as stated in the IEC 50160; within +10%/-15% of nominal voltage for 100% of the time. In Europe this translates in a range of 196 – 253 Volts. If the voltage gets out of this range, the charging may be stopped. Recently, because of the increasing voltage in the grid, due to the increase in local renewable energy production like photovoltaic cells, there are talks about increasing the overvoltage protection. The overvoltage protection defines the voltage at which electricity producing units are allowed to disconnect from the grid. There is now a hard limit at 253V, but this will be made into a 10 minute average value with a hard limit of +15% or 264.5V.

Because the height of the grid voltage is brought down by the current drawn (f.i. by DC fast chargers) and brought up by the current produced (f.i. by solar panels) ElaadNL recommends to remain charging if the voltage is too high, as long as it is within the specification of the OBC. When the voltage is too low, ElaadNL recommends to stop charging. The lower limit of -10% or 207 Volts can be used, which the grid needs to comply to for 95% of the time, or the -15% limit, which the grid needs to comply to for 100% of the time. This translates into a lower limit of 195.5 Volts. It is worth noting that these values are 10 minutes averages, so a short drop below 195.5 Volts might occur. We strongly advice to stop charging during these short voltage drops. When the voltage is within limits again, the charging should resume.

During these tests ElaadNL will also check if the vehicle compensates the changes in power because of the changes in the voltage by adapting the height of the current. This is tested on nominal charging speed and on the lowest charging speed, to see if the vehicle does not cross the maximum allowed current by the station in any situation.

5.1.1 Phase independent under voltage

It is possible the voltage sometimes is lower than desired. This can happen on all phases at once but also on 1 independent phase. During this test the voltage on the phases will be brought down in steps of 5 Volts till 195 Volt and in one step back to 230 Volts. It is tested how the vehicle responds when the voltage is lower and if it stops charging at a certain level. This can be tested on all phases independently from each other.

Estimated test time: 1 hour.

5.1.2 Phase independent over voltage

It is also possible the voltage sometimes is higher than desired. This can also happen on all phases at once or on 1 independent phase. ElaadNL can test how the vehicle responds when the voltage is higher and test if the vehicle will stop the charging (taken into account not damaging the vehicle). This can be tested on all phases independently from each other. During this test the voltage is increased in steps of 5 Volts till 260 Volt and in one step back to 230 Volts.

Estimated test time: 1 hour.

5.1.3 Phase voltage unbalance

A voltage on one phase can be higher or lower than on the other. During this test the voltage on one phase can be set to the lower limit, while the other phase is set to the upper limit. Different combinations are made to check the immunity to any scenario. This test is only applicable for multi-phase charging vehicles.

Estimated test time: 10 minutes.

5.1.4 Voltage sags

It is possible the voltage sags/dips for a short amount of time. At ElaadNL it is possible to test if the vehicle responds to a such a sag by shortly dropping the voltage to a lower level (100 Volts for 30 cycles). This can be done phase independently.

Estimated test time: 20 minutes.

5.2. Grid frequency test

The electricity grid in Europa normally operates at a frequency of close to 50 Hz. This means the current on each phase describes a full sinus wave 50 times per second. When demand and supply are not stable, this frequency can deviate. The IEC 50160 standard states that the grid voltage should always be within 47 Hz and 52 Hz. Because the frequency is generated by the rotating of the generators these will always be the same for the different phases.

5.2.1 *Under frequency*

It is possible the frequency sometimes is lower than desired. ElaadNL can test how the vehicle responds when the grid frequency is lower (47 Hz). If the vehicle stops charging, Elaad can test at what specific frequency the vehicle will stop the charging. If the vehicle remains charging, ElaadNL can lower the frequency further to find where the vehicle stops. This will only be done in accordance with the EV manufacturer.

Estimated test time: 30 minutes.

5.2.2 *Over frequency*

It is possible the frequency sometimes is higher than desired. ElaadNL can test how the vehicle responds when the grid frequency is higher (52 Hz). If the vehicle stops charging, Elaad can test at what specific frequency the vehicle will stop the charging. If the vehicle remains charging, ElaadNL can increase the frequency further to find where the vehicle stops. This will only be done in accordance with the EV manufacturer.

Estimated test time: 30 minutes.

5.3. Grid harmonic test

This is a test of charging behaviour when harmonics are added to the voltage according to the IEC 61000-4-13 standard. The EV is considered a class 2 product, see description below in figure 6. The ability of the vehicle to keep charging is tested (ride through) and the measurements are analysed for any unexpected changes in the charging currents due to the voltage harmonics.

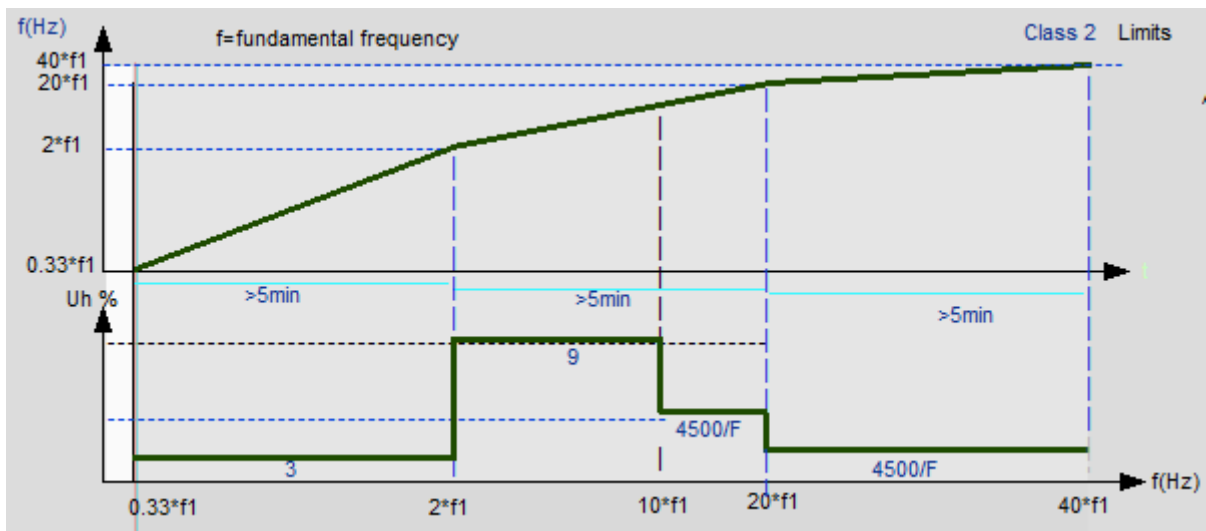
Class 2

This class applies to points of common coupling (PCC's for consumer systems) and in-plant points of common coupling (IPC's) in the industrial environment in general. The compatibility levels in this class are identical to those of public networks; therefore components designed for application in public networks may be used in this class of industrial environment.

Figure 5: definition of class 2 equipment 61000-4-13

The following scenario's will be tested:

5.3.1 Meister Curve



Our grid has a base frequency of 50Hz. Which means the meister curve can be divided into 4 blocks. Below is a table which summarises picture above of the harmonics induced during this test.

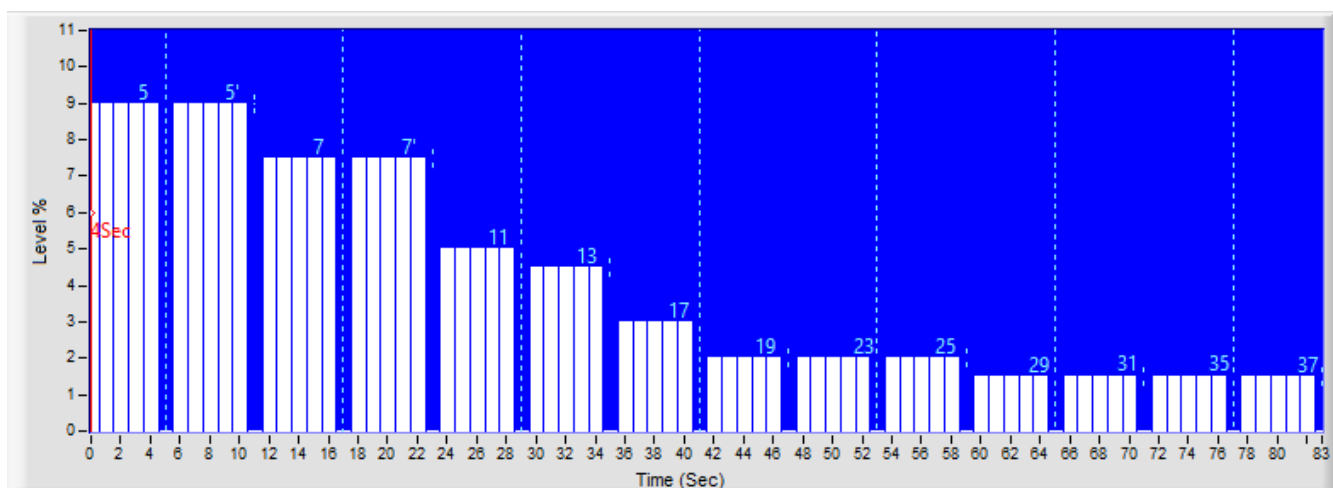
	Block 1	Block 2	Block 3	Block 4
From	16,5Hz	100Hz	500Hz	1000Hz
To	100Hz	500Hz	1000Hz	2000Hz
Intensity	3%	9%	(4500/current frequency)%	(4500/current frequency)%

Estimated test time: 1 hour.

5.3.2 Odd harmonics non multiple of 3

During the odd harmonics non multiple of 3 test the following harmonics are induced after each other:

5 th order	5 th order 180°	7 th order	7 th order 180°	11 th order	13 th order	17 th order	19 th order	23 rd order	25 th order	29 th order	31 st order	35 th order	37 th order
--------------------------	----------------------------------	--------------------------	----------------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------

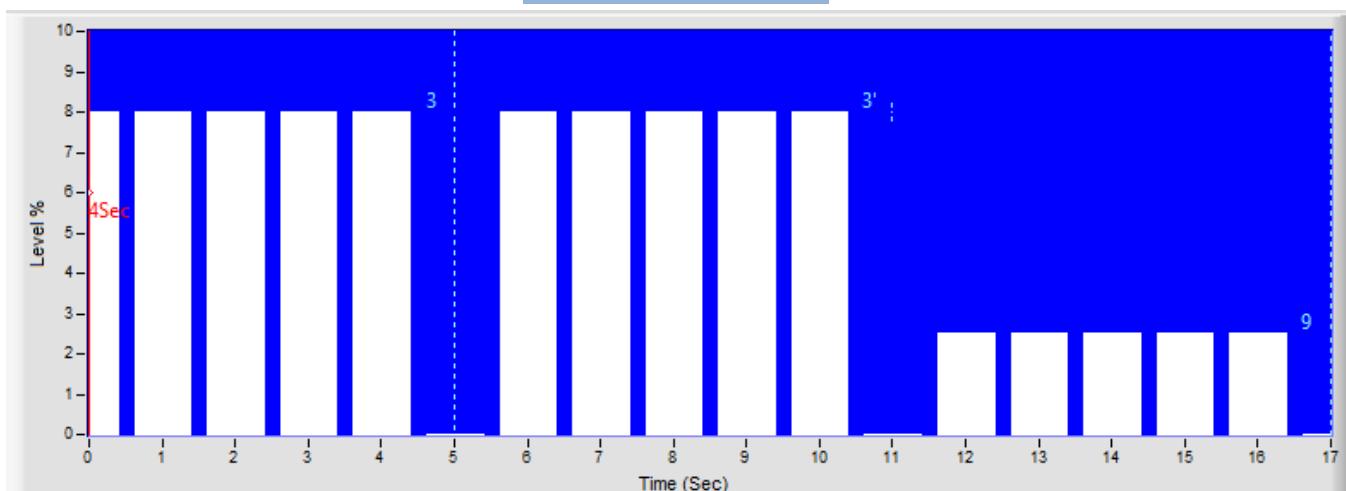


Estimated test time: 10 minutes.

5.3.3 Odd harmonics multiple of 3

During the odd harmonics multiple of 3 test the following harmonics are induced after each other:

3 rd order	3 rd order 180°	9 th order
--------------------------	----------------------------------	--------------------------

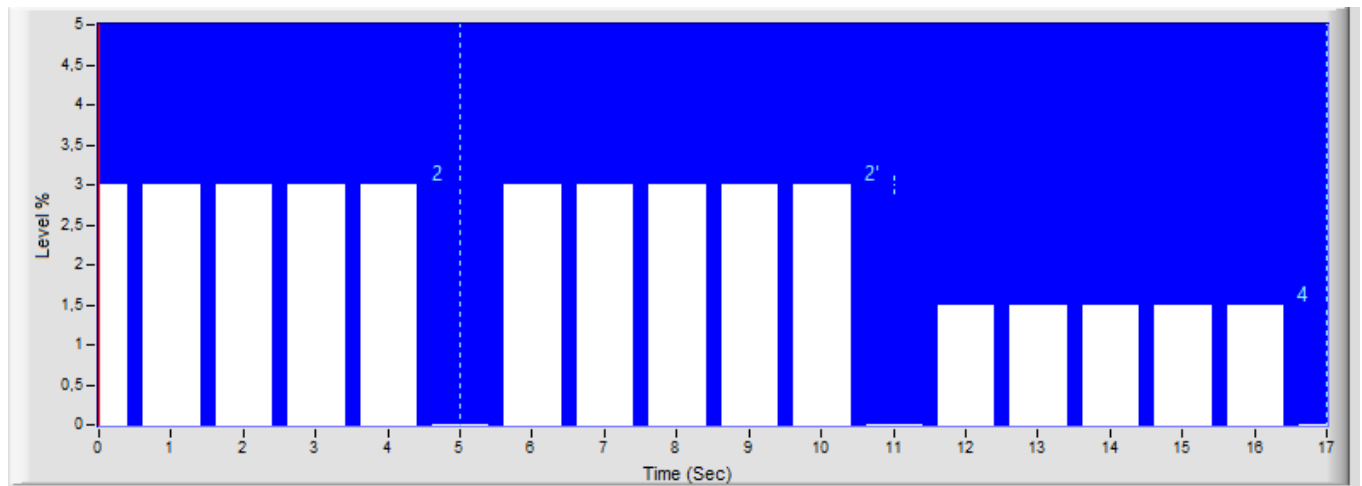


Estimated test time: 10 minutes.

5.3.4 Even harmonics

During the even harmonics test the following harmonics are induced after each other:

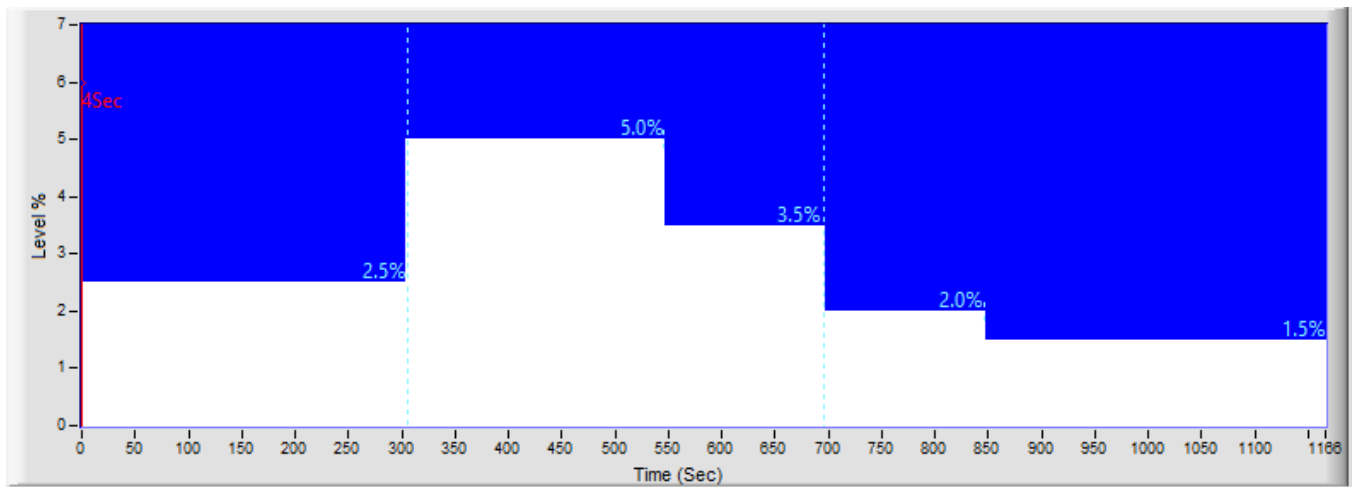
2 nd order	2 nd order 180°	4 th order
--------------------------	----------------------------------	--------------------------



Estimated test time: 10 minutes.

5.3.5 Interharmonics

During the interharmonics test the following harmonics are induced after each other:



For 50Hz mains For 60Hz mains

Class 2 Class 3 User Define

16-100Hz	=	2.5 %
100-500Hz	=	5 %
500-750Hz	=	3.5 %
750-1000Hz	=	2 %
1000-2000Hz	=	1.5 %

Estimated test time: 1,5 hours.

Annex A: Vehicle behaviour

Date:
Vehicle:
Test engineer(s):

Vehicle details:

License plate	
Brand	
Type	
Variant	
Version	
Trade name	
Type approval number	

This procedure is done once at one single charging station, to find out which events will trigger the vehicle to start or stop charging, if a certain plug-in order is expected by the vehicle and if the cable is locked during charging. In (at least) step 5 and 9 the vehicle is expected to start charging. The acquired information will be used to expand the general knowledge on vehicle behaviour to help customer to charge their vehicles and, most importantly, test if the cable is always locked at the vehicle during charging to prevent possible unsafe situations.

1. Vehicle cable plug-in and locking behaviour overview

Step	Action	Waiting time	Vehicle charging?	Cable Locked during charge?	Remarks
1	Unlock vehicle, Switch contact off				
2	Plug-in cable in vehicle				
3	Plug-in cable in charging station				
4	Activate by swiping card on RFID reader	30 sec			
5	Lock vehicle	30 sec	Yes? Go to 6 No? Change plug-in order and retry		
6	Unlock vehicle	30 sec (if applicable)			
7	Open door	30 sec			
8	Close door	30 sec			
9	Lock vehicle	30 sec			
10	Swipe card to stop charging				
11	Unplug vehicle				

12	Change plug in order (if not already done in step 5)				
----	--	--	--	--	--

2. Opening of S2 when cable is unlocked

Is the cable at the EV still powered (S2 still closed) when the cable is unlocked and ready to be pulled out?	Yes/No?
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3. Cable lock during power outage

Is the cable at the EV still locked when the charging station is powered off?	Yes/No?
---	---------

4. Lid lock

Can the lid be opened when the vehicle is locked?	Yes/No?
---	---------

5. Charging without ground connection?

Type of charger	Charging possible?	Remarks
Mode 3		
Mode 2		

6. Vehicle charging speed at different number of phases (only multi-phase vehicles)

Step	Action	Max current per phase?	Remarks
1	Connect station at three phase 32A		
2	Start transaction at 32A		
3	Stop transaction		
4	Connect station at 1 phase 32A		
5	Start transaction at 32A		
6	Stop transaction		

7. Vehicle behaviour with N-E (de)coupled in test set-up

N-E Coupled?	Charging possible?	Remarks
Yes		
No		

8. Vehicle stop by reaching predefined charge level.

Does the vehicle open S2 and does it stay open?	Yes/No?
---	---------

9. Total absence of voltage

ID	Requirement	Expected result	Observations
a	Check vehicle response to PWM without AC being delivered during charging	Observation Test	
b	Check vehicle response to PWM without AC being delivered at start of charging	Observation Test	

10. Phase independent absence of voltage at start transaction (only on multi-phase vehicles)

Absence of voltage					
Voltage absent on phase	Able to charge?		No voltage present on non-powered phases?	self-healing when voltage is added again?	Issues / Remarks
	2 phase EV	3 phase EV			
L1					
L2					
L3					
L1 + L2					
L2 + L3					
L1 + L3					

11. Phase independent absence of voltage during a transaction (only on multi-phase vehicles)

Absence of voltage					
Voltage absent on phase:	Able to charge?		No voltage present on non-powered phases?	self-healing when voltage is added again?	Issues / Remarks
	2 phase EV	3 phase EV			
L1					
L2					
L3					
L1 + L2					
L2 + L3					
L1 + L3					

12. Phase independent low voltage at L2 and L3 at start transaction (only on multi-phase vehicles)

Low voltage					
Voltage on L2 and L3 at start	Able to charge?		No phase-linking?	self-healing when voltage is brought back to 230V	Issues / Remarks
	2 phase EV	3 phase EV			
150					

Vehicle

100					
50					

13. Description of vehicle's behaviour

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14. Final remarks and comments

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Annex B: Test form Interoperability

Date:
Vehicle:
Test engineer(s):

1. Overview of stations and results

1.1 DC charging stations

Vendor	Type	Connector Type	Regular charging	Smart charging	Vehicle stops
Ambibox	Ambicharge	CCS			
Autel	DC Fast	CCS			
Autel	DC Wall Mount	CCS			
Delta	EVDE25E	CCS			
eNovates	10kW DC V2G	Chademo			
Heliox	40kW mobile	CCS			
Heliox	Rapid 150	CCS			
Heliox	Flex 180	CCS			
Ingeteam	Ingerev rapid 180	CCS			
Kostal	DC 11kW	CCS			
Nio	Nio Power	CCS			
Vestel	EVC06	CCS			
Vestel	EVC03	CCS			
Remarks					

1.2 (Semi) Public AC charging stations

Vendor	Type	Connector Type	Mode 3	Smart charging	Vehicle stops
Alfen	EVE double pro-line	Type2			
Alfen	Tube 2.0	Type2			
Alfen	Twin XL	Type2			
Alfen	Twin 4XL	Type2			
Ecotap	Yeris wide 64A	Type2			
eNovates	Pro11	Type2			
eNovates	Pro22	Type2			
EV-Box	Business line	Type2			
EV-Box	Elvi	(Fixed) Type2			
EV-Box	ML zuil 64A	Type2			
GE	Durastation (LMS)	Type2			
Reewoud	Chargepoint P6 V2G-LMS	Type2			
Smappee	EV Base	Type 2			
Remarks					

1.3 Wallboxes

Vendor	Type	Connector Type	Mode 3	Smart charging	Vehicle stops
Autel	MaxiCharger AC Wallbox	Type 2			
CTEK	Chargestorm Connected 2	Type 2			
Easee	Home	Type 2			
EO Charging	Genius 2	Type 2			
Etrell	Inch Home	Type 2			
Liteon	EU 32A intelligent charger	Type 2			
Schneider Electric	EVLink smart wallbox	Type 2			
Schneider Electric	EVLink Pro	Type 2			
Smapppee	EV Wall	Type 2			
Vestel	EVC04	Type 2			
Wallbox	Copper SB	Type 2			
Wallbox	Commander 2	(Fixed) Type 2			
Remarks					

2. Interoperability test cases

2.1 Mode 3 interoperability (per station)

Step	Action	Mode 3 voltage	OK?	Remarks
1	Plug-in cable in charging station	12		
2	Plug-in cable in vehicle	9		
3	Start transaction on station	9		
4	PWM signal starting	9		
5	Vehicle closes S2	6		

2.2 Smart charging interoperability (per station, according (time) table below)

Step	Action	Ok?	Remarks
1	Set up charging session		
2	Start charging profile		
3	Current change high-low		
4	Short charge interrupt		
5	Long charging interrupt		

Profile with short interrupt

Start time (s)	0	30	60	90	120	150	180	480
Current (A)	25	6	25	0	6	25	0	16
Current (PWM)	41,7	10	41,7	100	10	41,7	100	26,7

Profile with long interrupt

Start time (s)	0	1200	1260	1320
Current (A)	0	6	8	16
Current (PWM)	100	10	13,3	26,7

2.3 Vehicle charge interrupted (per station)

Test	Method	Does vehicle open S2?	Does vehicle restart and close s2?	Remarks
1	A: Stop by delayed charging			
2	B: Stop at vehicle side (e.g. unlock vehicle)			

2.4 Interoperability with different protocols

Protocol	Charging possible?	Remarks
ISO15118-2		
DIN70121		

2.5 AC 15118-2 test form

Testcase	OK	Issues / Remarks
General communication setup		
Unsecure smart charging		
Secure communication (TLS only)		
Smart charging		
Plug & Charge + Smart Charging		
Plug & Charge + Certificate install		
Secure communication (TCP & TLS)		
Smart charging		
Plug & Charge + Smart Charging		
Plug & Charge + Certificate install		
Secure communication (Private Environment)		

3. Final remarks and comments

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Annex C: Test form Smart Charging

Date:
Vehicle:
Test engineer(s):

1. Smart charging timings

Current PWM	Current RMS	Overshoot?	Within 5s?	Within 3s?	S2 open within 3s?	remarks
0						
6						
9						
12						
16						
20						
24						
28						
32						
28						
24						
20						
16						
12						
9						
6						
0						
32						
6						
32						
0						

2. Smart charging timings (1 Phase)

Current PWM	Current RMS	Overshoot?	Within 5s?	Within 3s?	S2 open within 3s?	remarks
0						

6						
9						
12						
16						
20						
24						
28						
32						
28						
24						
20						
16						
12						
9						
6						
0						
32						
6						
32						
0						

3. Smart charging overview

ID	Requirement	OK	Issues / Remarks
1a	Fluctuating charging profile		
1b	Fluctuating with no charge in between		
1b	Fluctuating with on normal grid		
2a	Time division 0A-32A		
2b	Time division 0A-6A		
3a	Delayed charging 30 min		
3b	Paused charging 20 min		
3c	6 hours delayed and paused		
3d	3 hours low charging speed		
4	Charging speed change after 3 hours		
5	PWM Signal Test (below 8% and above 97% PWM, no charging is allowed)		

6	Check if the vehicle keeps charging at max and min limit when the accessory circuits are active.		
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4. Final remarks and comments

NOTE: *If the end user can use an informational app, it is advised to clearly indicate the difference between charging is pausing and charging has stopped.*

Annex D: Power quality emissions to Grid

Date:
Vehicle:
Test engineer(s):

1. PQ emissions 1 phase charging

Current PWM	Current RMS	THD	First 40 harmonics	Supra-harmonics	Cos Phi	Total PF	Issues / Remarks
6							
9							
12							
16							
20							
24							
28							
32							
28							
24							
20							
16							
12							
9							
6							

2. PQ emissions 3 phase charging

Current PWM	Current RMS	THD	First 40 harmonics	Supra-harmonics	Cos Phi	Total PF	Issues / Remarks
6							

Vehicle

9							
12							
16							
20							
24							
28							
32							
28							
24							
20							
16							
12							
9							
6							

3. Overall results

ID	Requirement	OK	Issues / Remarks
1	THD		
2	1-40th Harmonic above EC-61000-3-2/12 norm?		
3	Noticeable Supraharmonic distortions (2-150 kHz)		
4	No low power factor during charging (PF >0,85)		
5	Inrush current/transients > 3 times nominal current		
6	Frequency beating (2 EVs)		

4. Final remarks and comments

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Vehicle

Annex E: Power Quality immunity: voltage

Date:
Vehicle:
Test engineer(s):

1. Undervoltage (From 230V to 200V in steps of 5V and jumping back to 230V) at Max and Min charging current

Undervoltage Test							
Max current	Undervoltage on phase	EV keeps charging?			EV compensates current?	EV restarts charging at 230V?	Issues / Remarks
		EV charging capability:					
		1 Phase	2 phases	3 phases			
Max (Vehicle dependent)	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						
6A	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						

2. Overvoltage (From 230V to 260V in steps of 5V and jumping back to 230V) at Max and Min charging current

Overvoltage Test							
Max current	Overvoltage on phase	EV keeps charging?			EV compensates current?	EV restarts charging at 230V?	Issues / Remarks
		EV charging capability:					
		1 Phase	2 phases	3 phases			
Max (Vehicle dependent)	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						
6A	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						

3. Phase voltage unbalance (only applicable for 2 or 3 phase vehicle chargers)

Phase voltage unbalance					
Charging speed	High (H), normal (N) and Low (L) Phase	EV keeps charging?	EV compensates current?	EV restarts charging at 230V?	Issues / Remarks
16A	L1(H) + L2(L) + L3 (N)				
	L1(H) + L2(N) + L3 (L)				
	L1(L) + L2(H) + L3 (N)				
	L1(L) + L2(N) + L3 (H)				

Vehicle

6A	L1(N) +L2(H) + L3 (L)				
	L1(N) +L2(L) + L3 (H)				
	L1(H) +L2(L) + L3 (N)				
	L1(H) +L2(N) + L3 (L)				
	L1(L) +L2(H) + L3 (N)				
	L1(L) +L2(N) + L3 (H)				
	L1(N) +L2(H) + L3 (L)				
	L1(N) +L2(L) + L3 (H)				

4. Voltage sag at Max and Min charging current

Voltage sag							
Max current	Volage sag on phase	EV keeps charging?			EV compensates current?	EV restarts charging at 230V?	Issues / Remarks
		EV charging capability:					
		1 Phase	2 phases	3 phases			
Max (Vehicle dependent)	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						
6A	L1						
	L2						
	L3						
	L1 + L2						
	L2 + L3						
	L1 + L3						
	L1 + L2 + L3						

Vehicle

5. Final remarks and comments

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Annex F: Power Quality immunity: frequency

Date:
Vehicle:
Test engineer(s):

1. Test of charging behaviour when in frequency range 47 – 52 Hz @ Maximum power and minimum power during 1 phase charging

	Freq. (Hz)	Charging?		Changes in current intake?	Issues / Remarks
		Max A	6A		
Max current (Vehicle dependent) / 6A	50				
	49				
	48				
	47				
	50				
	51				
	52				

2. Test of charging behaviour when in frequency range 47 – 52 Hz @ Maximum power and minimum power during 3 phase charging (if applicable)

	Freq. (Hz)	Charging?		Changes in current intake?	Issues / Remarks
		Max A	6A		
Max current (Vehicle dependent) / 6A	50				
	49				
	48				
	47				
	50				
	51				
	52				

3. Overall results

ID	Requirement	OK	Issues / Remarks
1	Under frequency		
2	Over frequency		

4. Final remarks and comments

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Annex G: Power Quality immunity: harmonics

Date:
Vehicle:
Test engineer(s):

1. Test of charging behaviour when adding harmonics to the voltage according to the IEC 61000-4-13 standard. The EV is considered a class 2 product. Test performed at Max and Min charging speed and

1a. Measurements during 1 phase charging

Test	Current in A	Ride through?	Unexpected changes in current intake or on PF	Issues / Remarks
Meister curve	6A			
	Max			
Odd harmonics non-multiple of 3	6A			
	Max			
Odd harmonics multiple of 3	6A			
	Max			
Even harmonics	6A			
	Max			
Interharmonics	6A			
	Max			

1b. Measurements during 3 phase charging (if applicable)

Test	Ride through?	Unexpected changes in current intake or on PF	Issues / Remarks
	A		
Meister curve			
Odd harmonics non-multiple of 3			
Odd harmonics multiple of 3			
Even harmonics			
Interharmonics			

2. Overall results

ID	Requirement	OK	Issues / Remarks
1	Ride through?		

Vehicle

2	Changes in current intake?		
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3. Final remarks and comments

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Annex H: measurement screenshots

1. General vehicle behaviour and safety

2. Smart Charging

3. Power quality emissions

4. Power quality immunity: voltage

5. Power quality immunity: harmonics